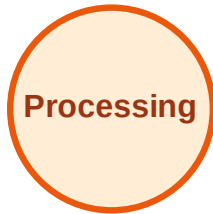


BFS Traversal

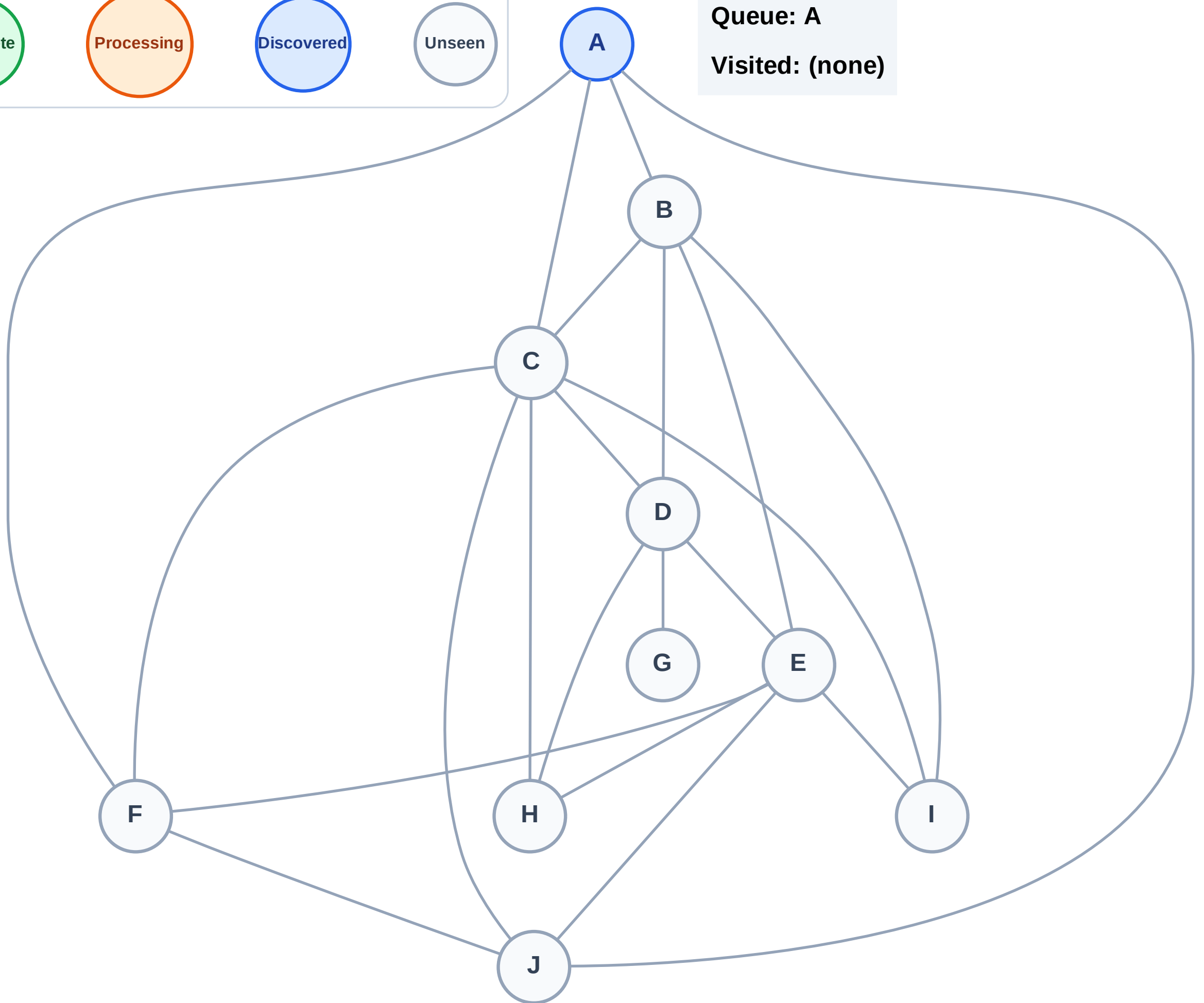
Step 1: Start at A

Legend



Queue: A

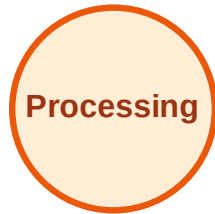
Visited: (none)



BFS Traversal

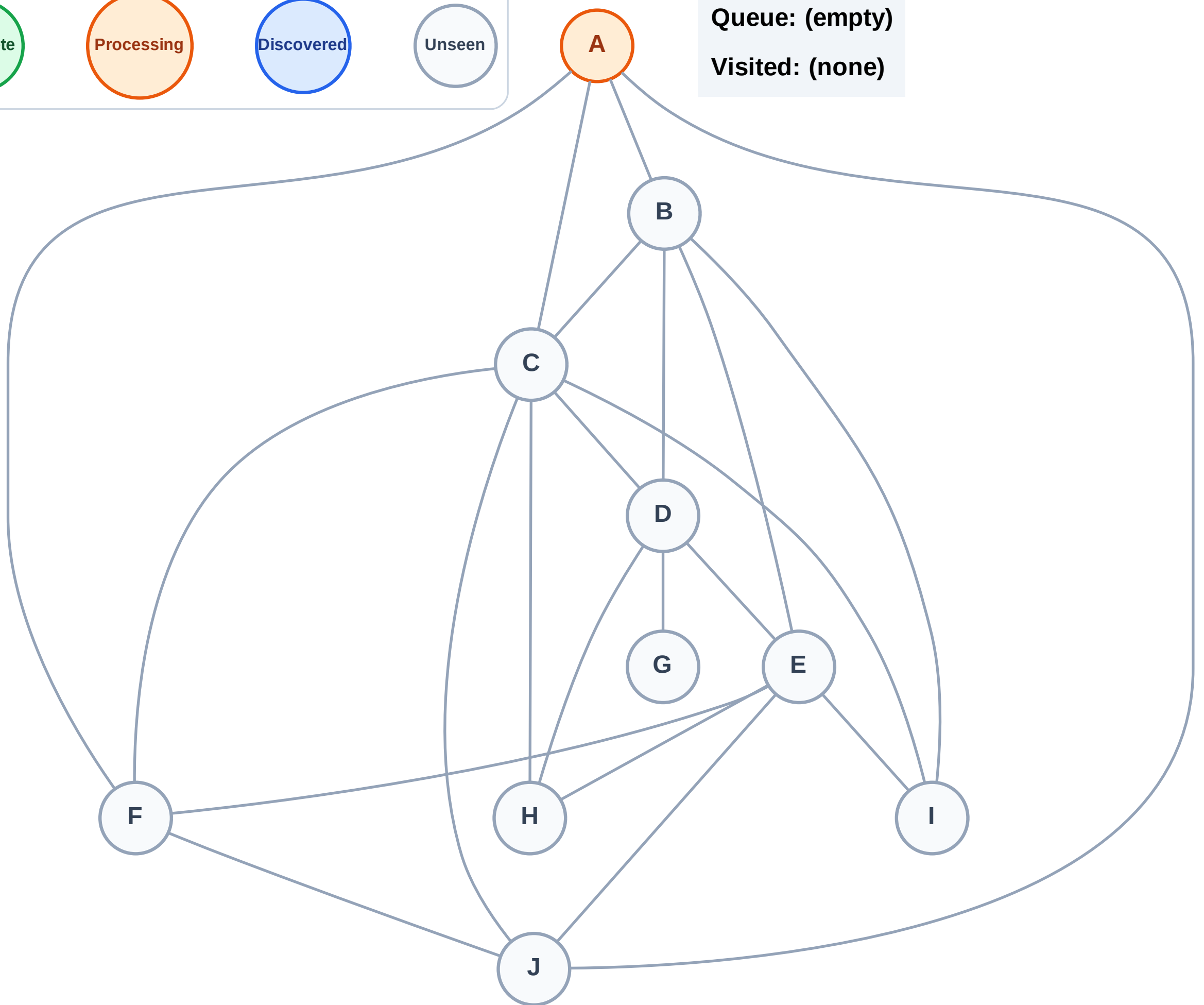
Step 2: Process node A

Legend



Queue: (empty)

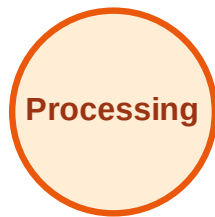
Visited: (none)



BFS Traversal

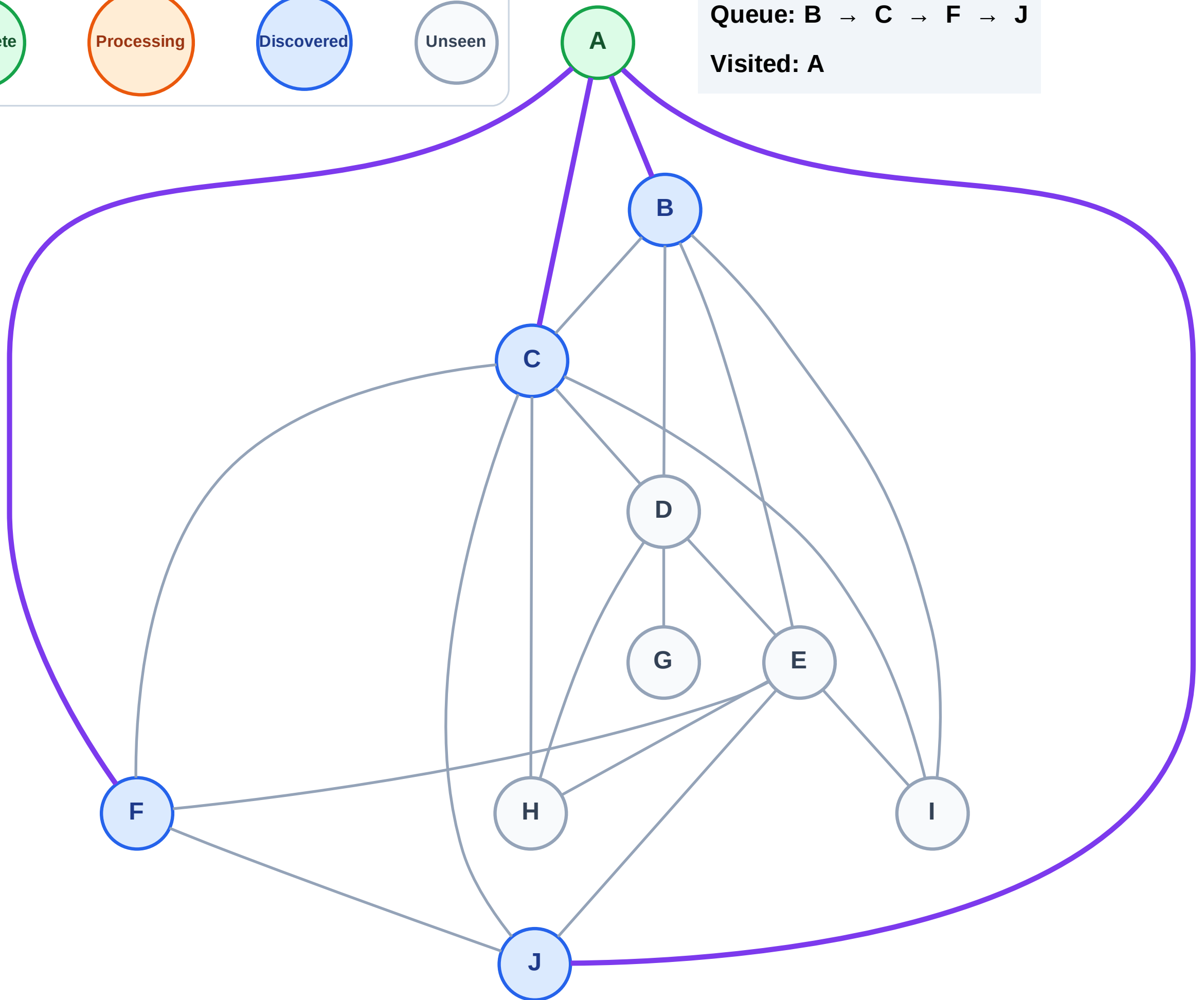
Step 3: Discover B, C, F, J from A

Legend



Queue: B → C → F → J

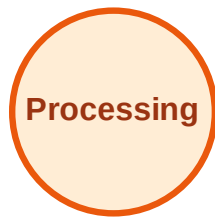
Visited: A



BFS Traversal

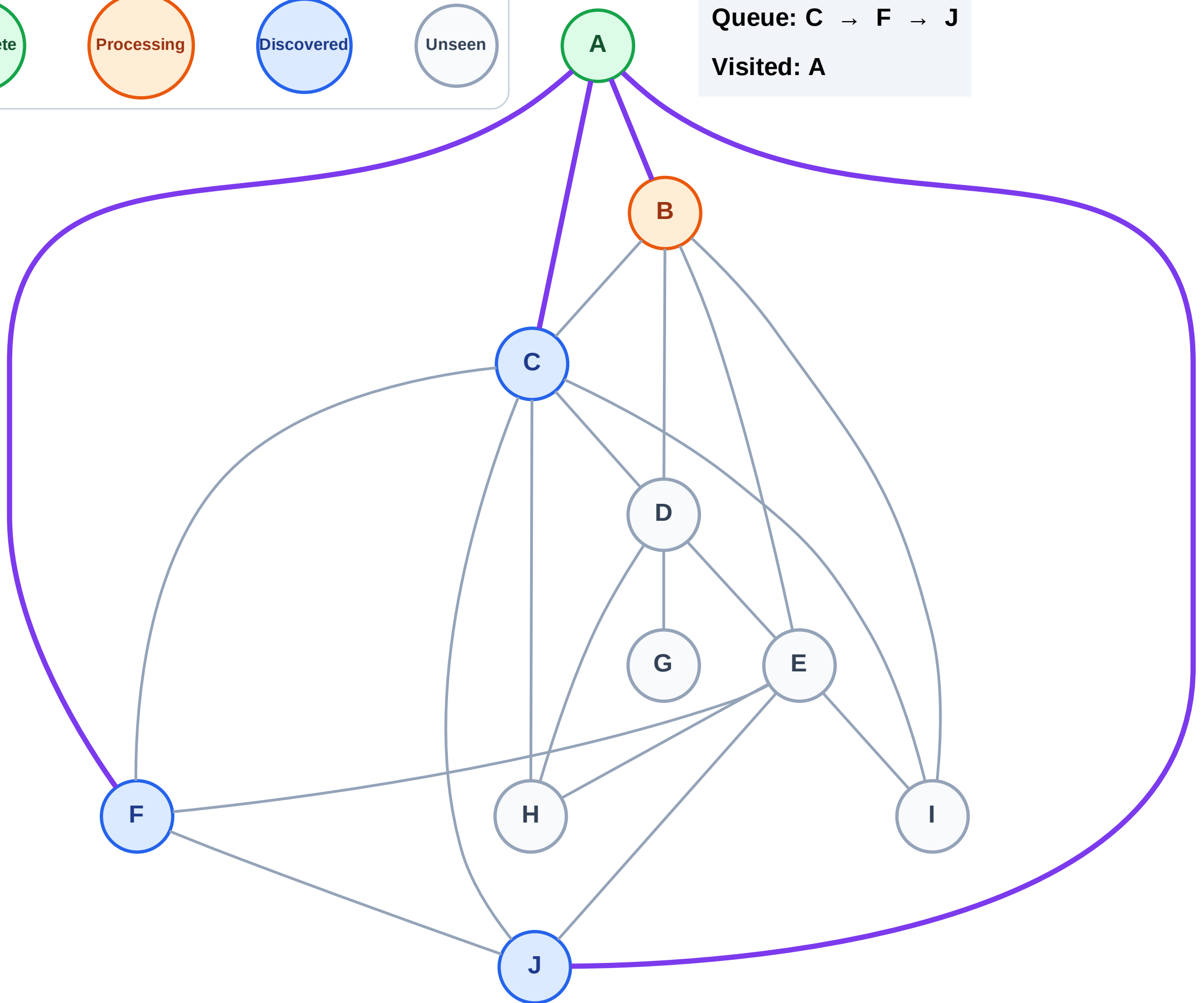
Step 4: Process node B

Legend



Queue: C → F → J

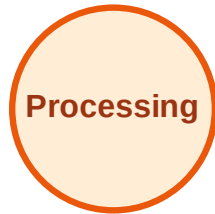
Visited: A



BFS Traversal

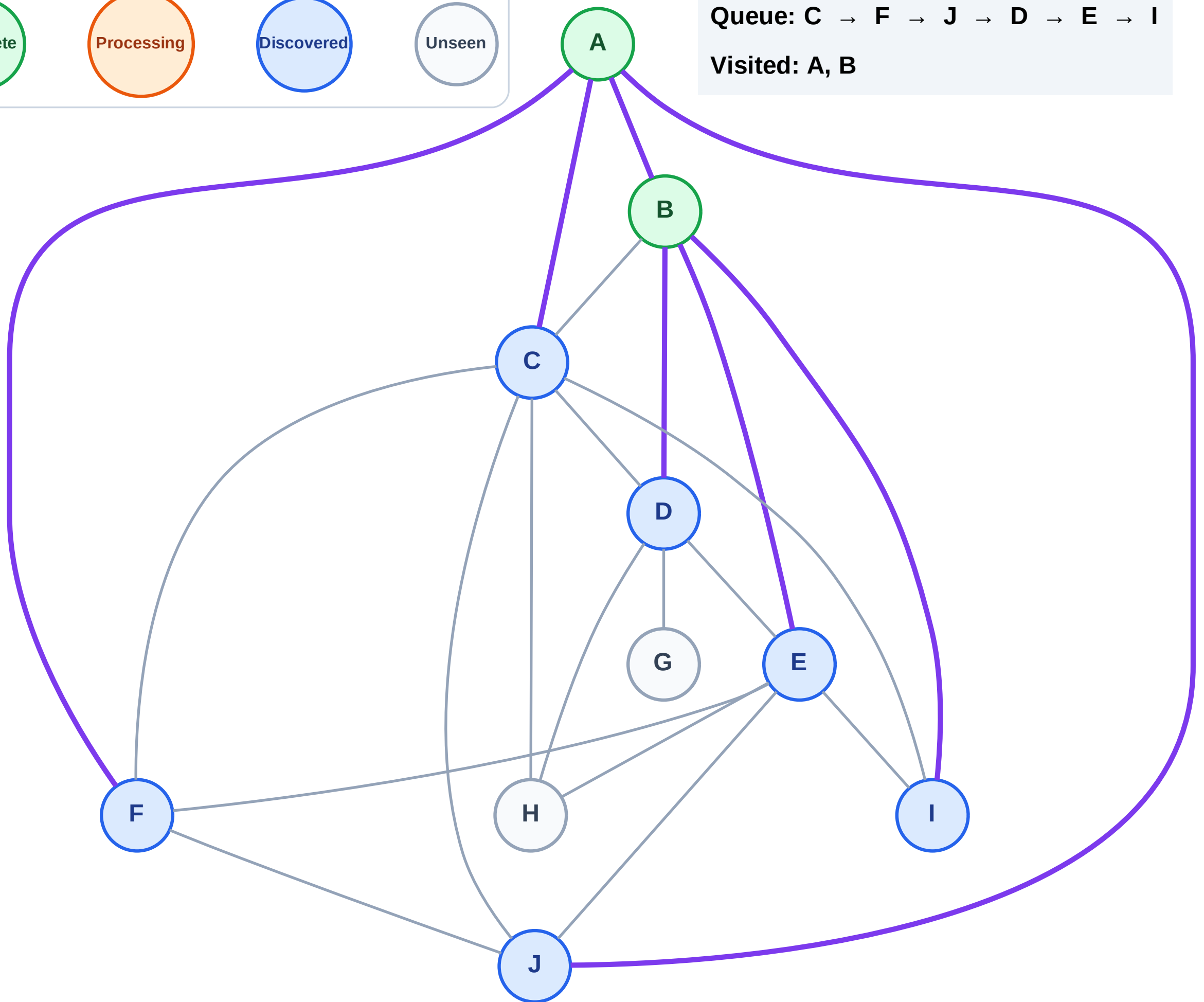
Step 5: Discover D, E, I from B

Legend



Queue: C → F → J → D → E → I

Visited: A, B



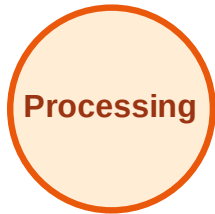
BFS Traversal

Step 6: Process node C

Legend



Complete



Processing



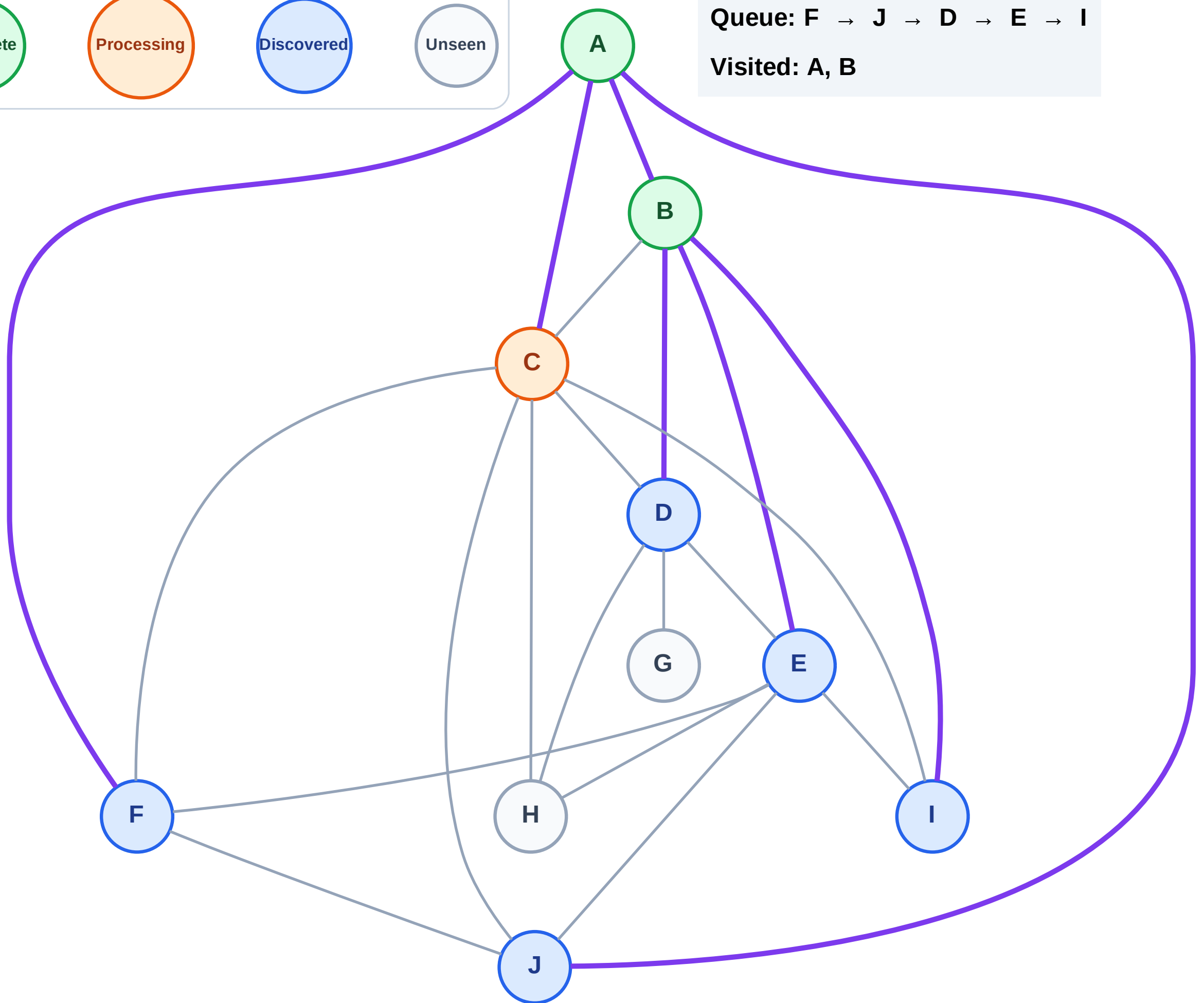
Discovered



Unseen

Queue: F → J → D → E → I

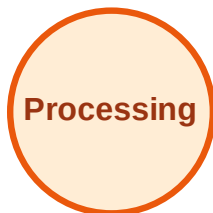
Visited: A, B



BFS Traversal

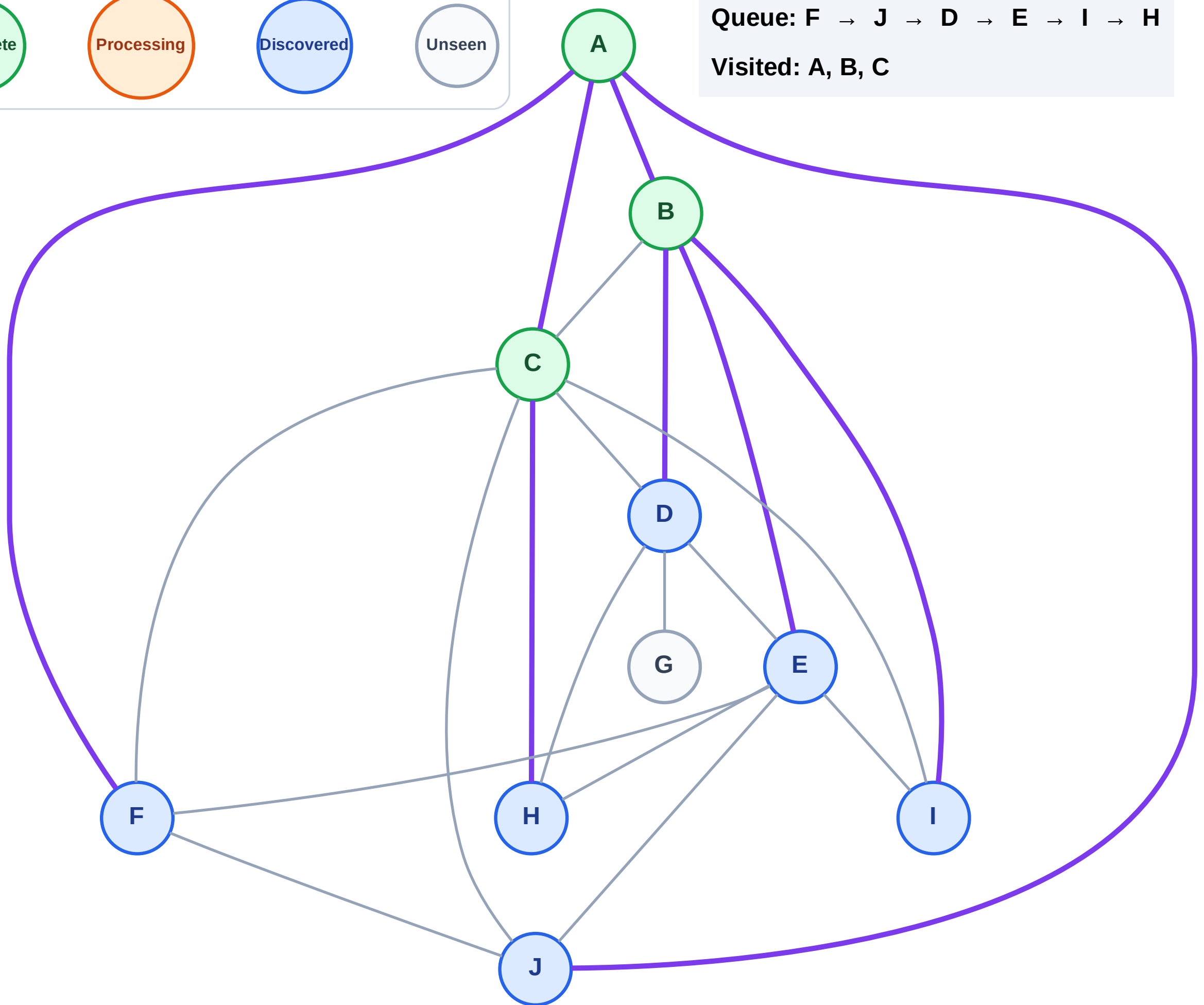
Step 7: Discover H from C

Legend



Queue: F → J → D → E → I → H

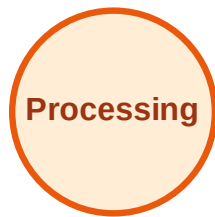
Visited: A, B, C



BFS Traversal

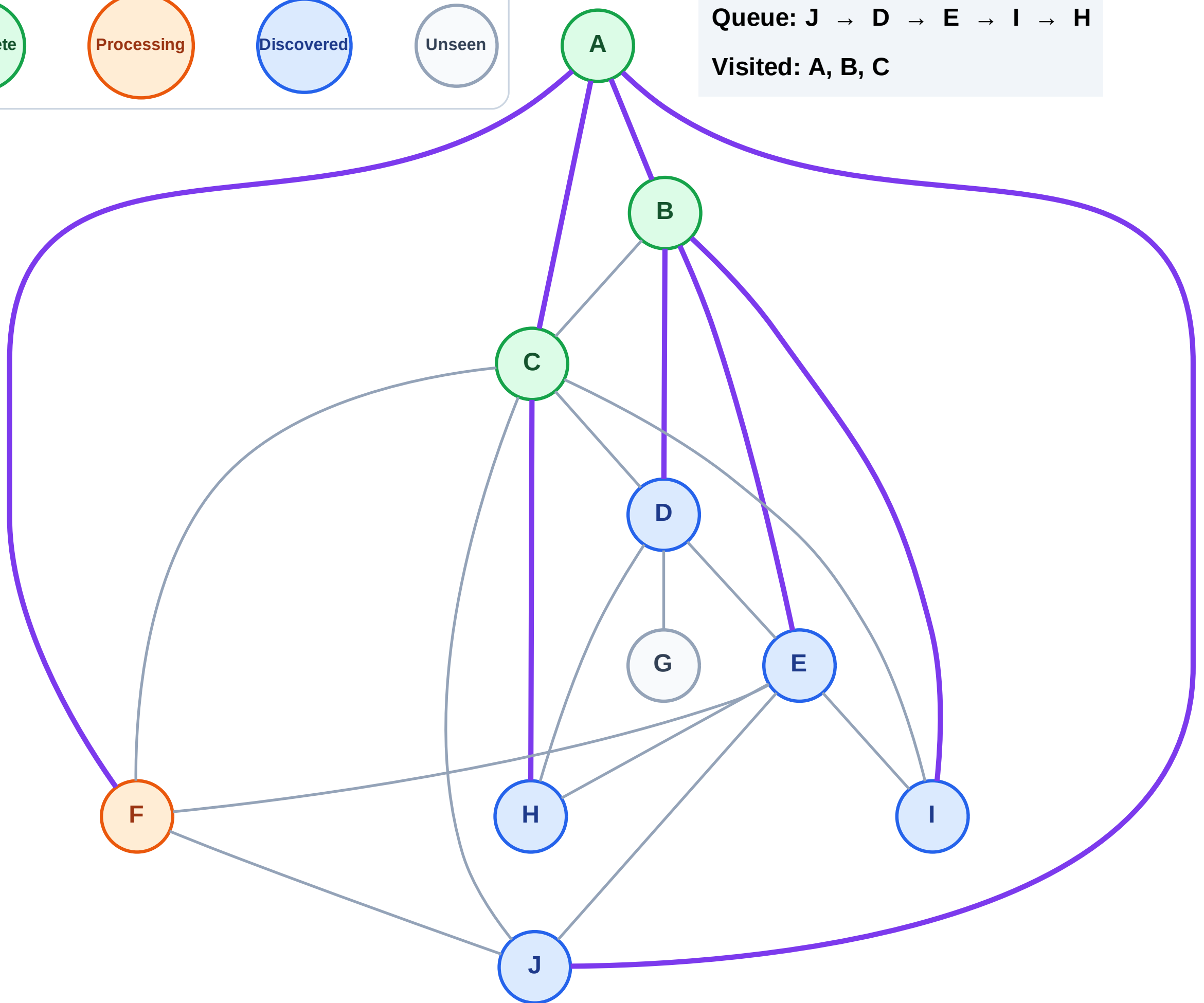
Step 8: Process node F

Legend



Queue: J → D → E → I → H

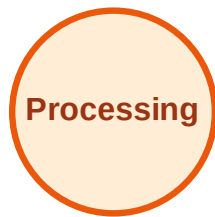
Visited: A, B, C



BFS Traversal

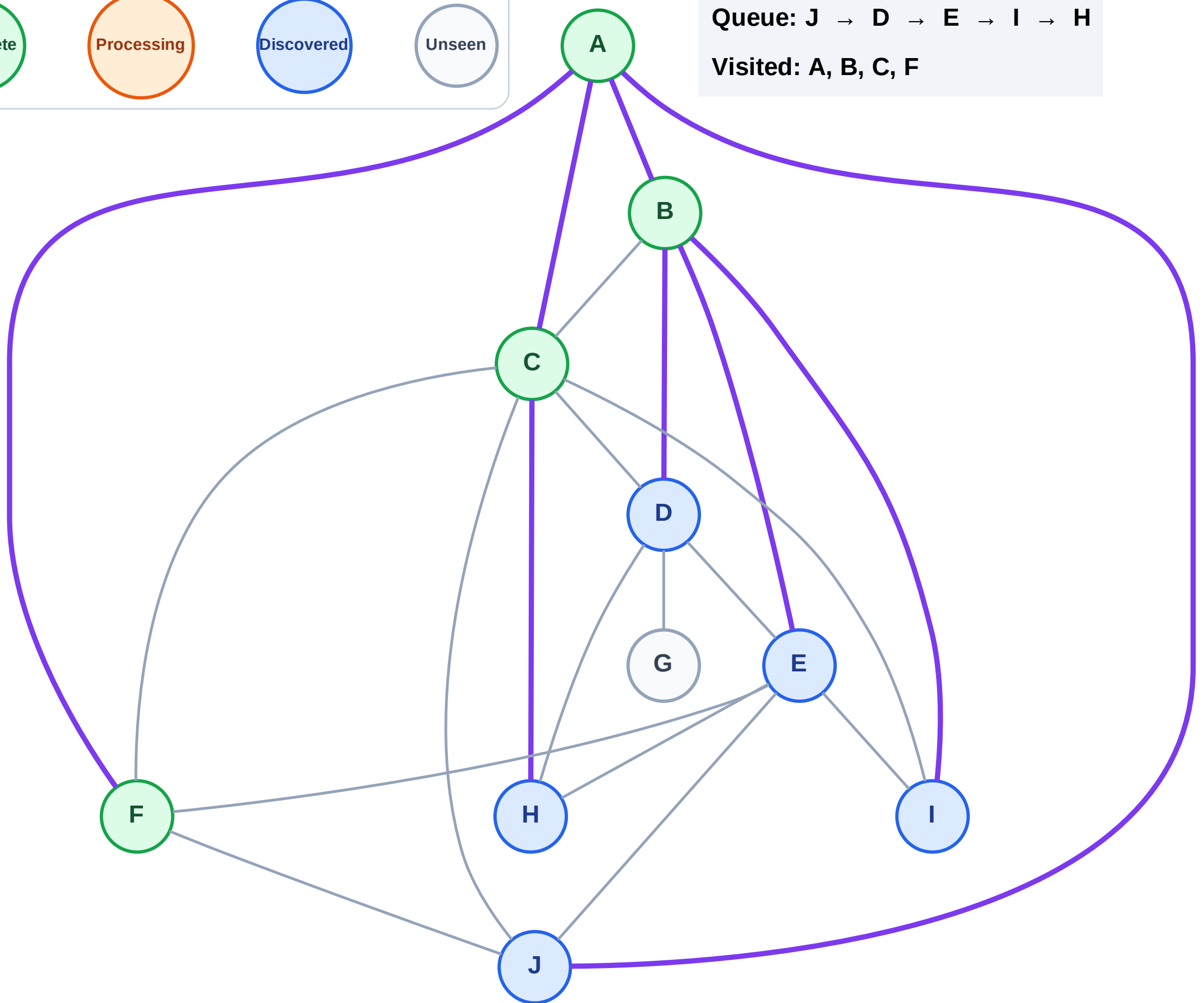
Step 9: No new neighbors from F

Legend



Queue: J → D → E → I → H

Visited: A, B, C, F



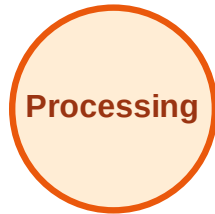
BFS Traversal

Step 10: Process node J

Legend



Complete



Processing



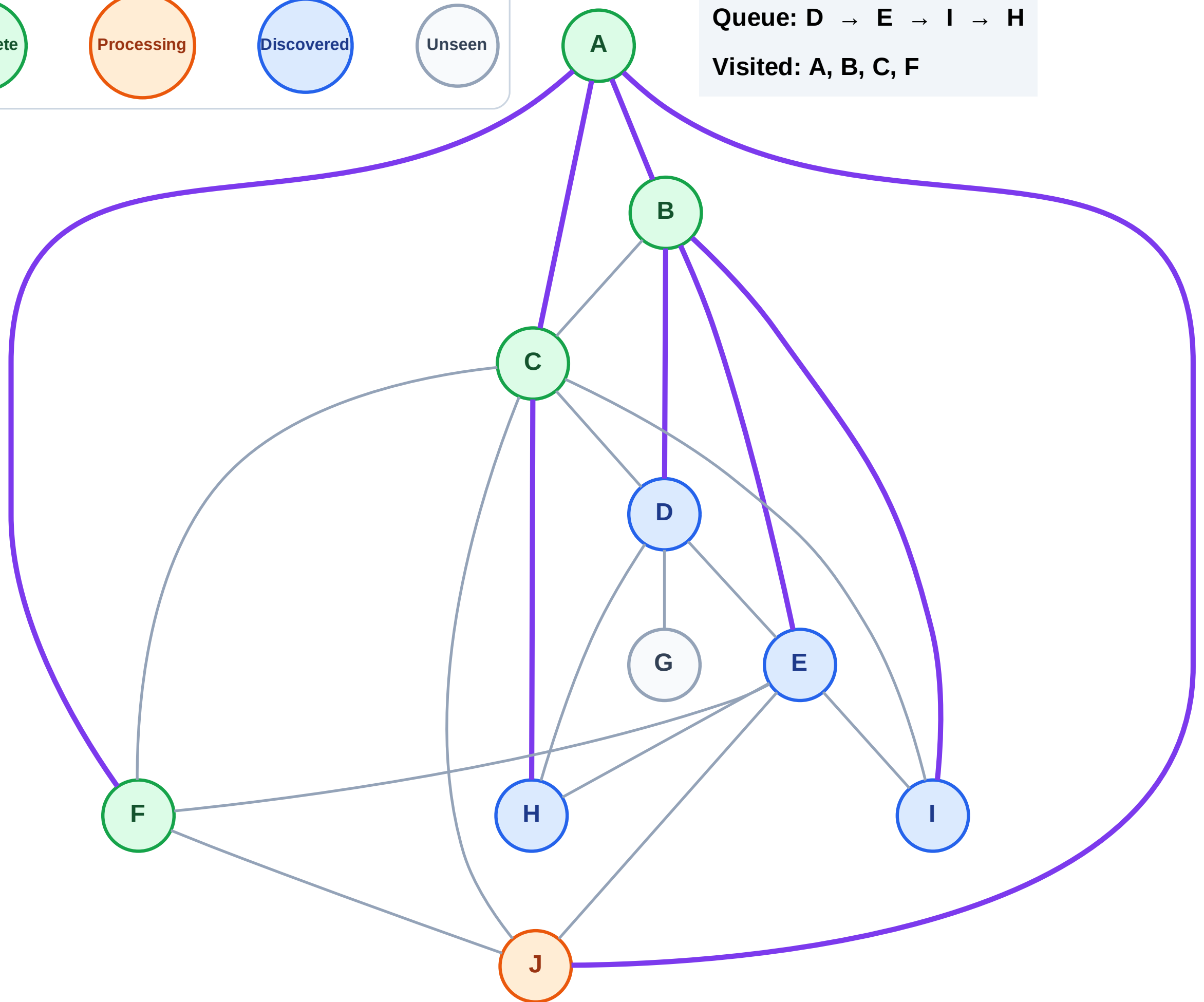
Discovered



Unseen

Queue: D → E → I → H

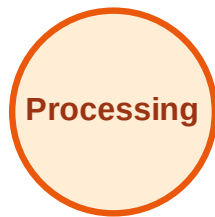
Visited: A, B, C, F



BFS Traversal

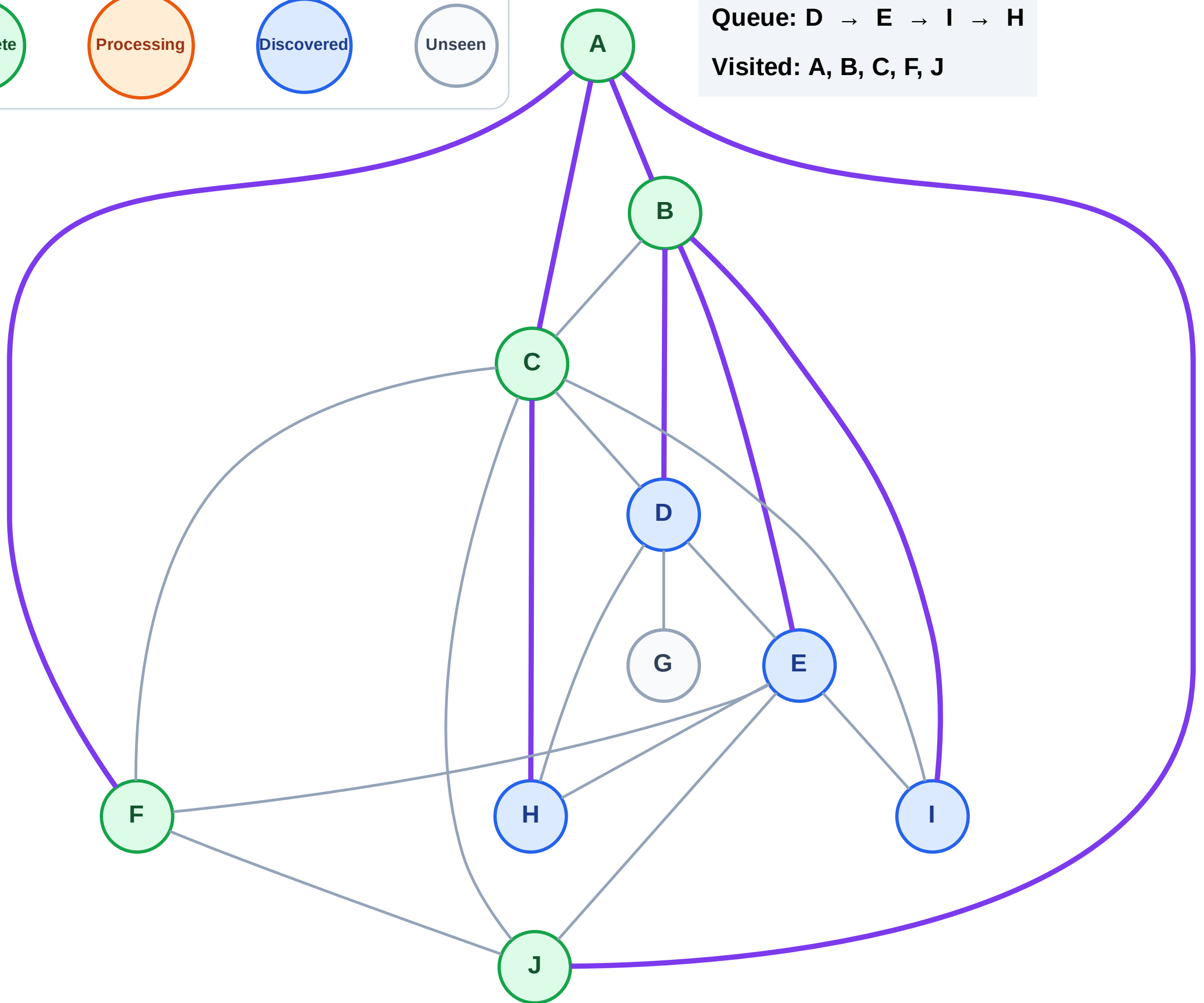
Step 11: No new neighbors from J

Legend

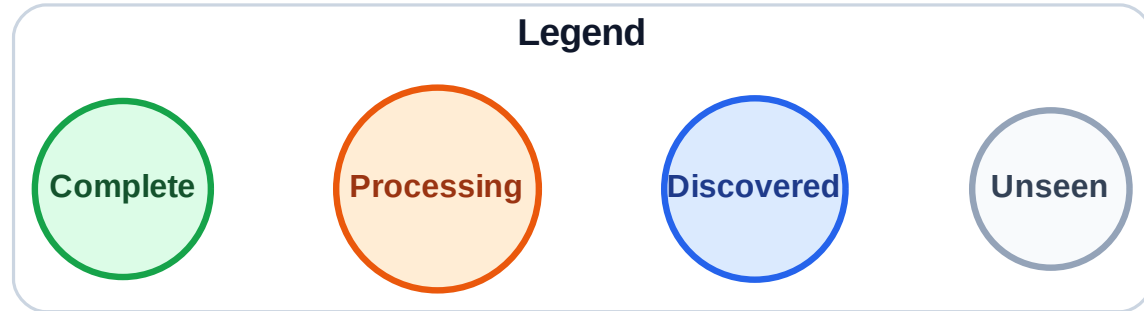


Queue: D → E → I → H

Visited: A, B, C, F, J

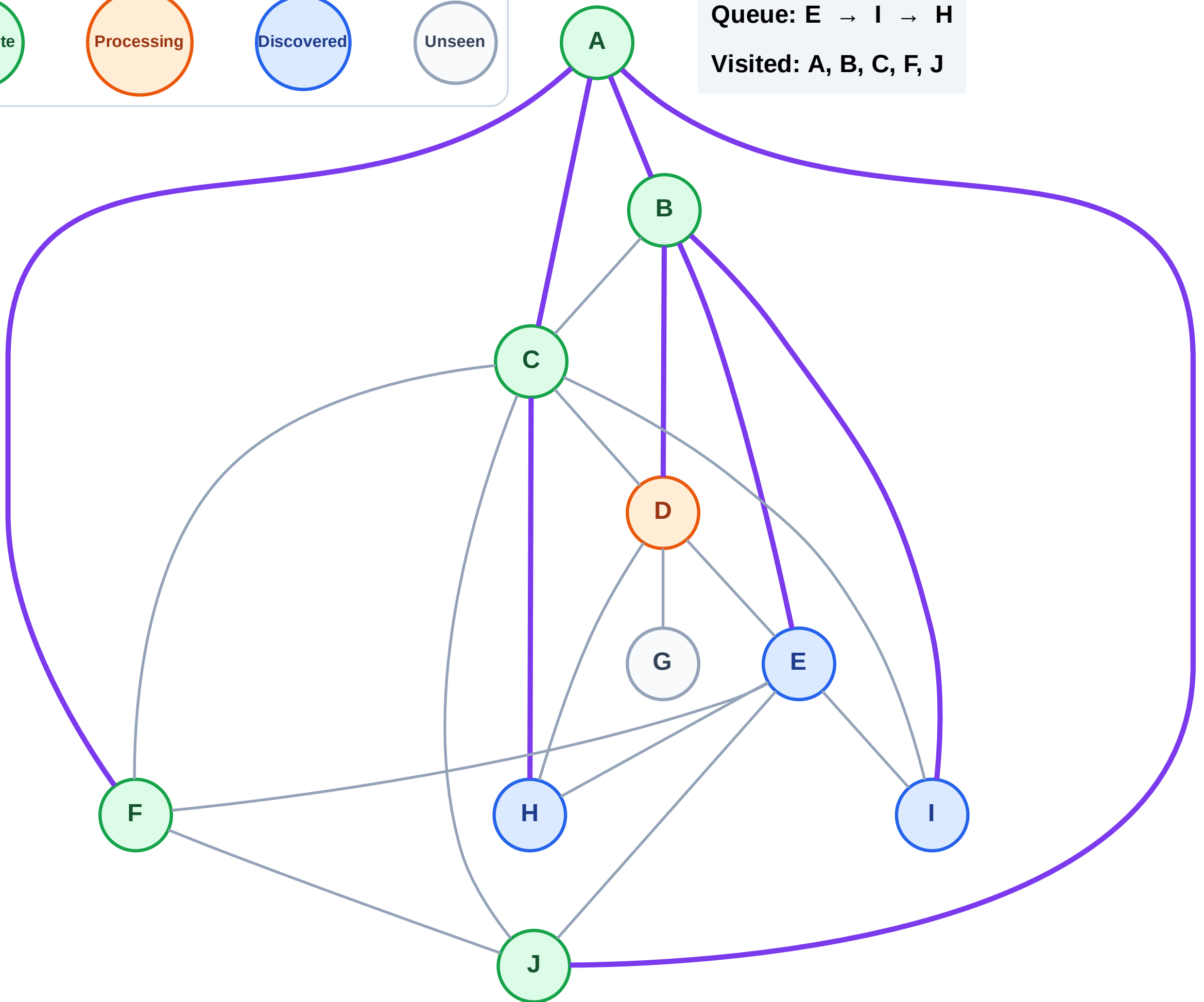


Step 12: Process node D



Queue: E → I → H

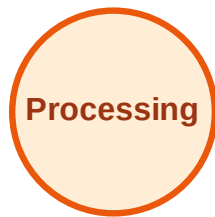
Visited: A, B, C, F, J



BFS Traversal

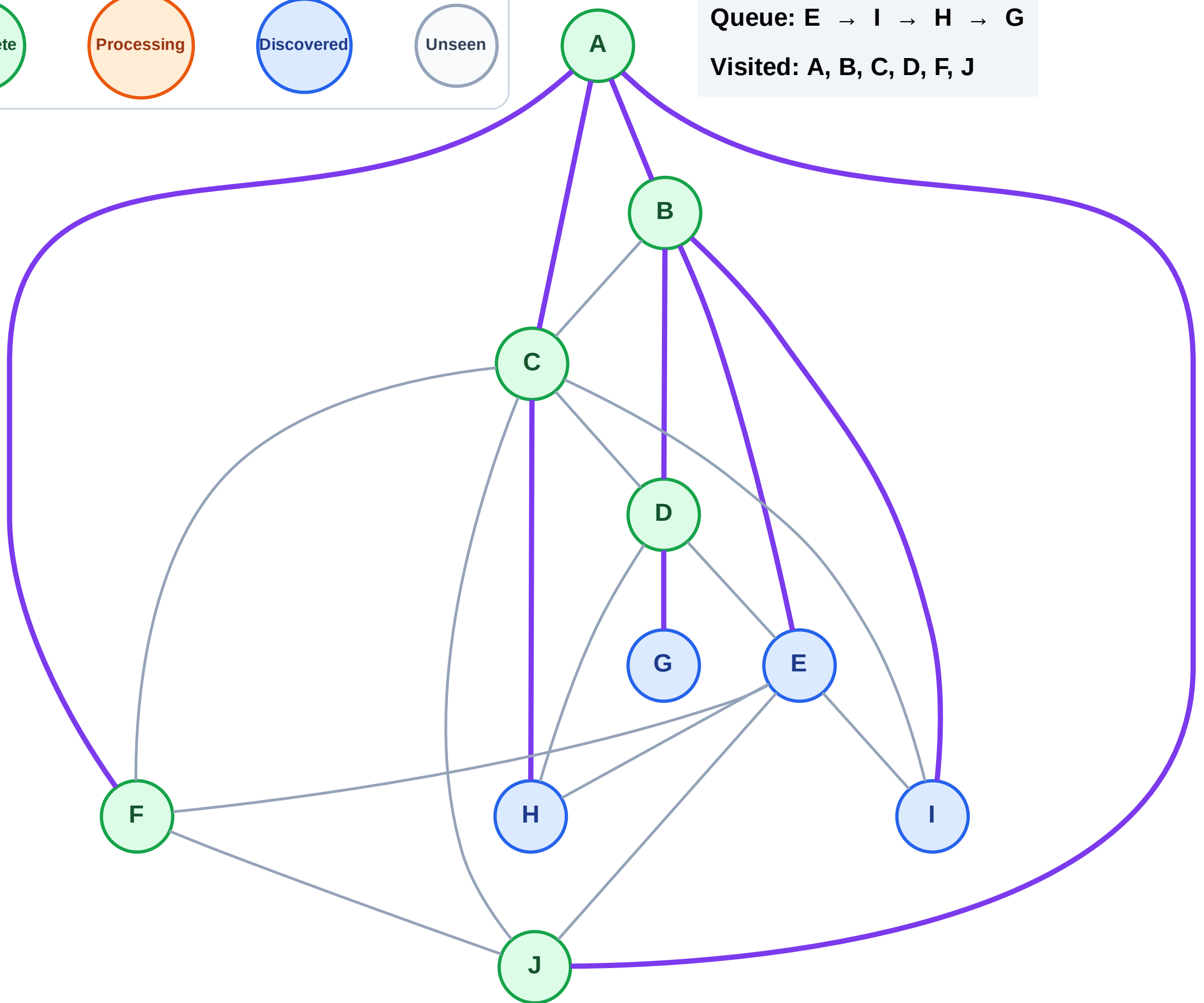
Step 13: Discover G from D

Legend



Queue: E → I → H → G

Visited: A, B, C, D, F, J



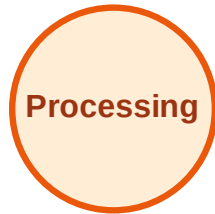
BFS Traversal

Step 14: Process node E

Legend



Complete



Processing



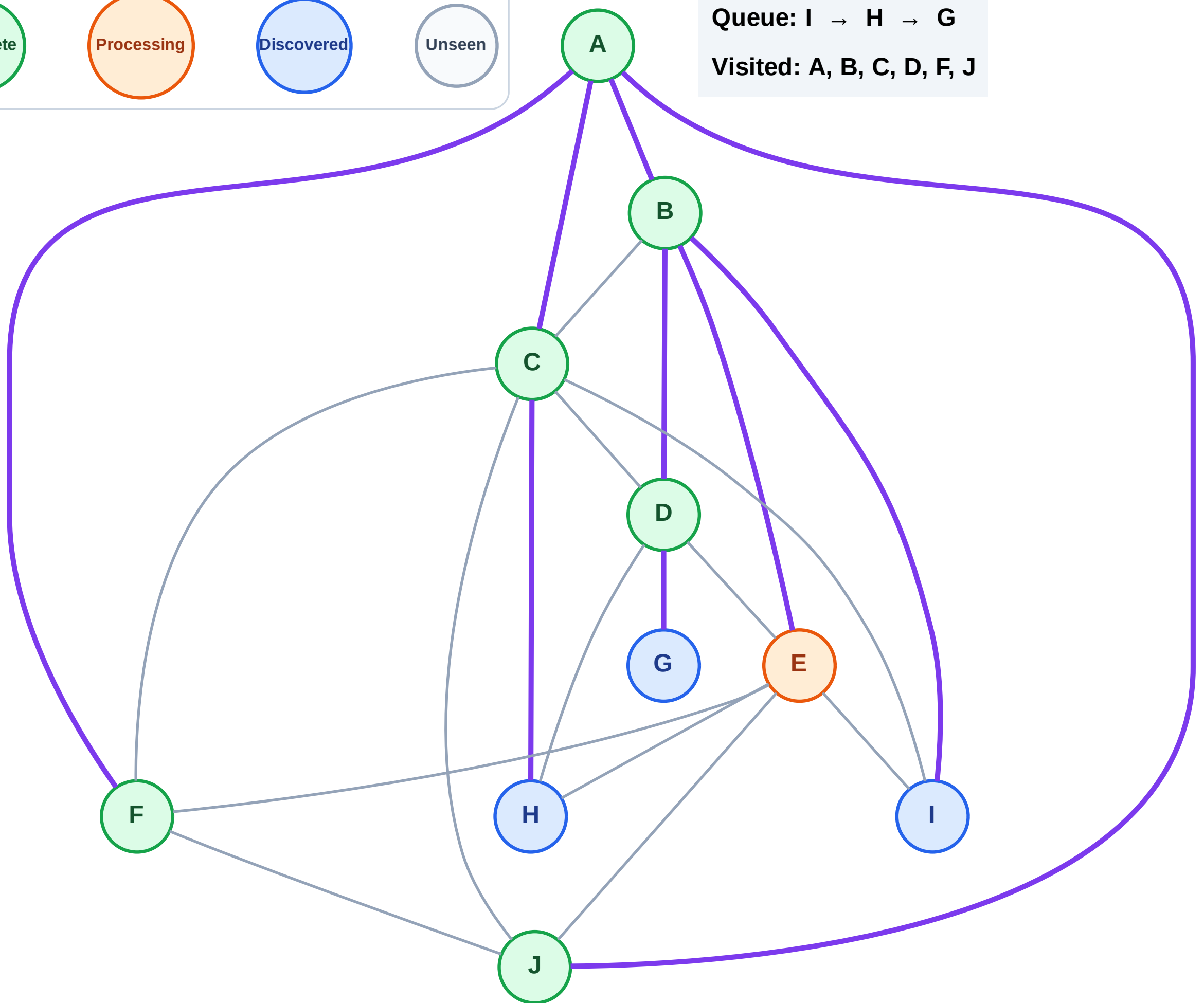
Discovered



Unseen

Queue: I → H → G

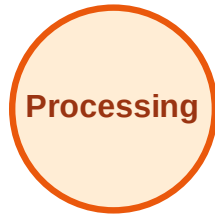
Visited: A, B, C, D, F, J



BFS Traversal

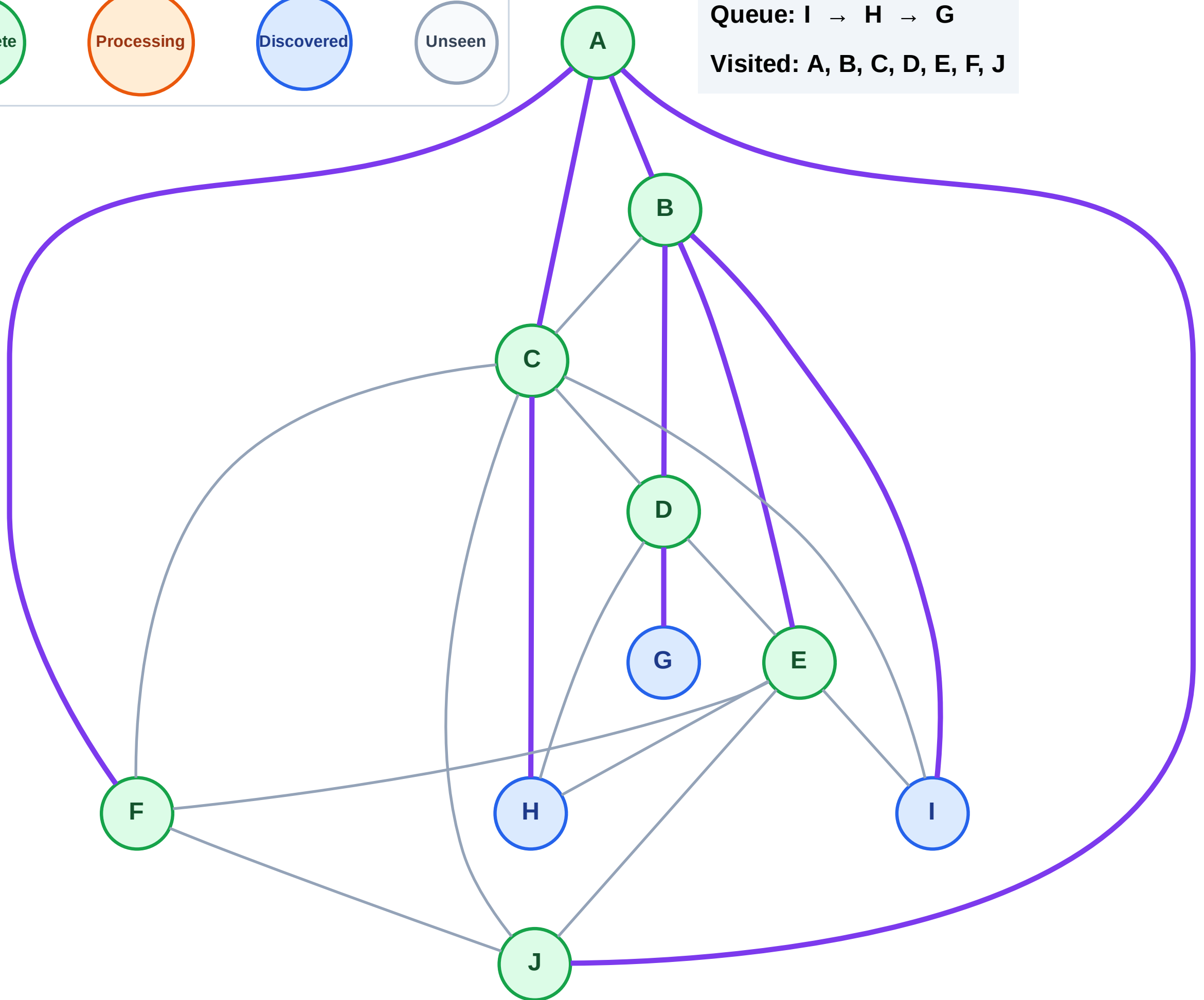
Step 15: No new neighbors from E

Legend



Queue: I → H → G

Visited: A, B, C, D, E, F, J



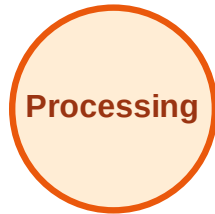
BFS Traversal

Step 16: Process node I

Legend



Complete



Processing



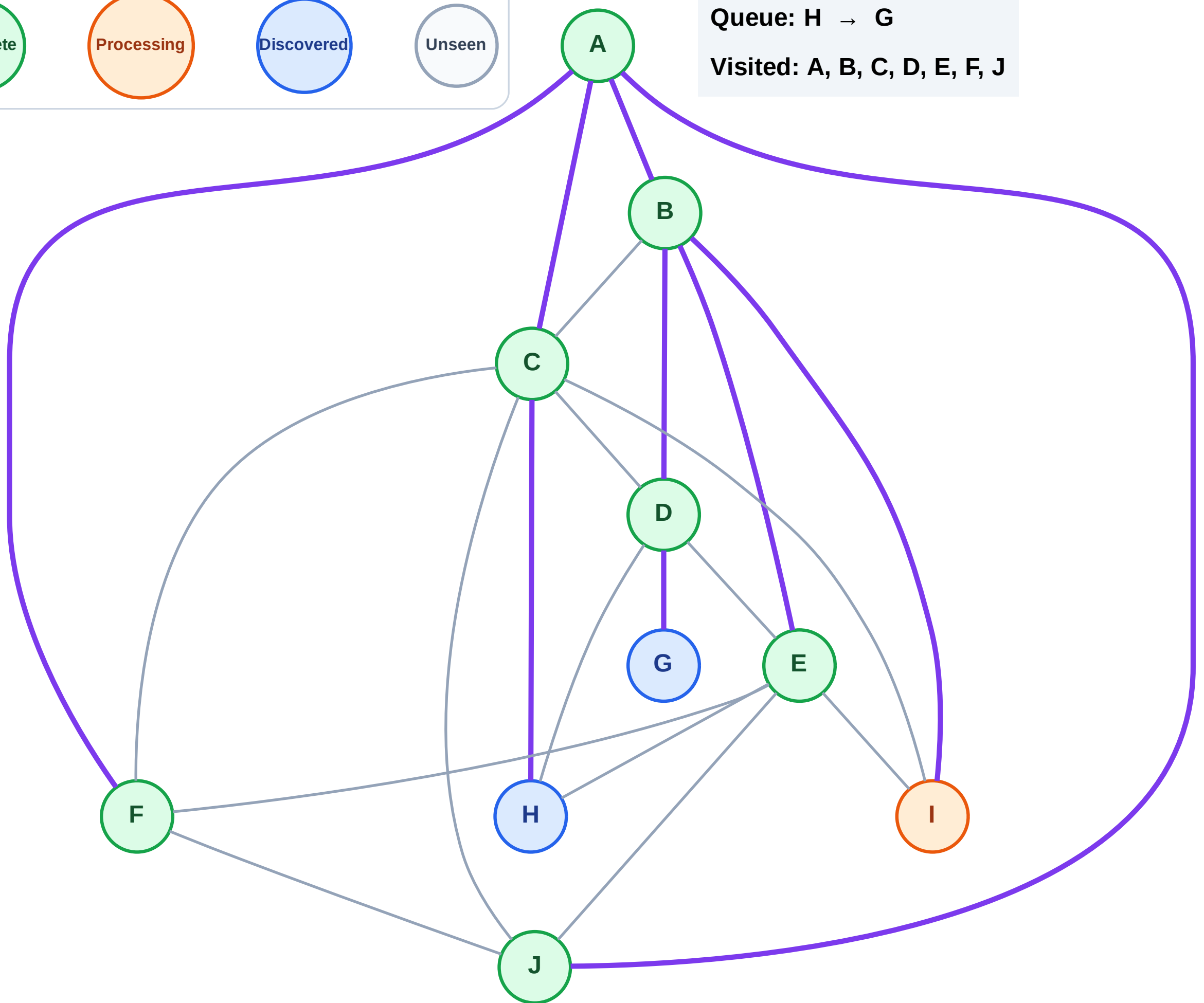
Discovered



Unseen

Queue: H → G

Visited: A, B, C, D, E, F, J



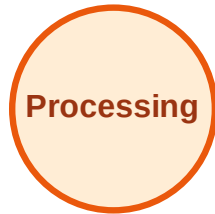
BFS Traversal

Step 17: No new neighbors from I

Legend



Complete



Processing



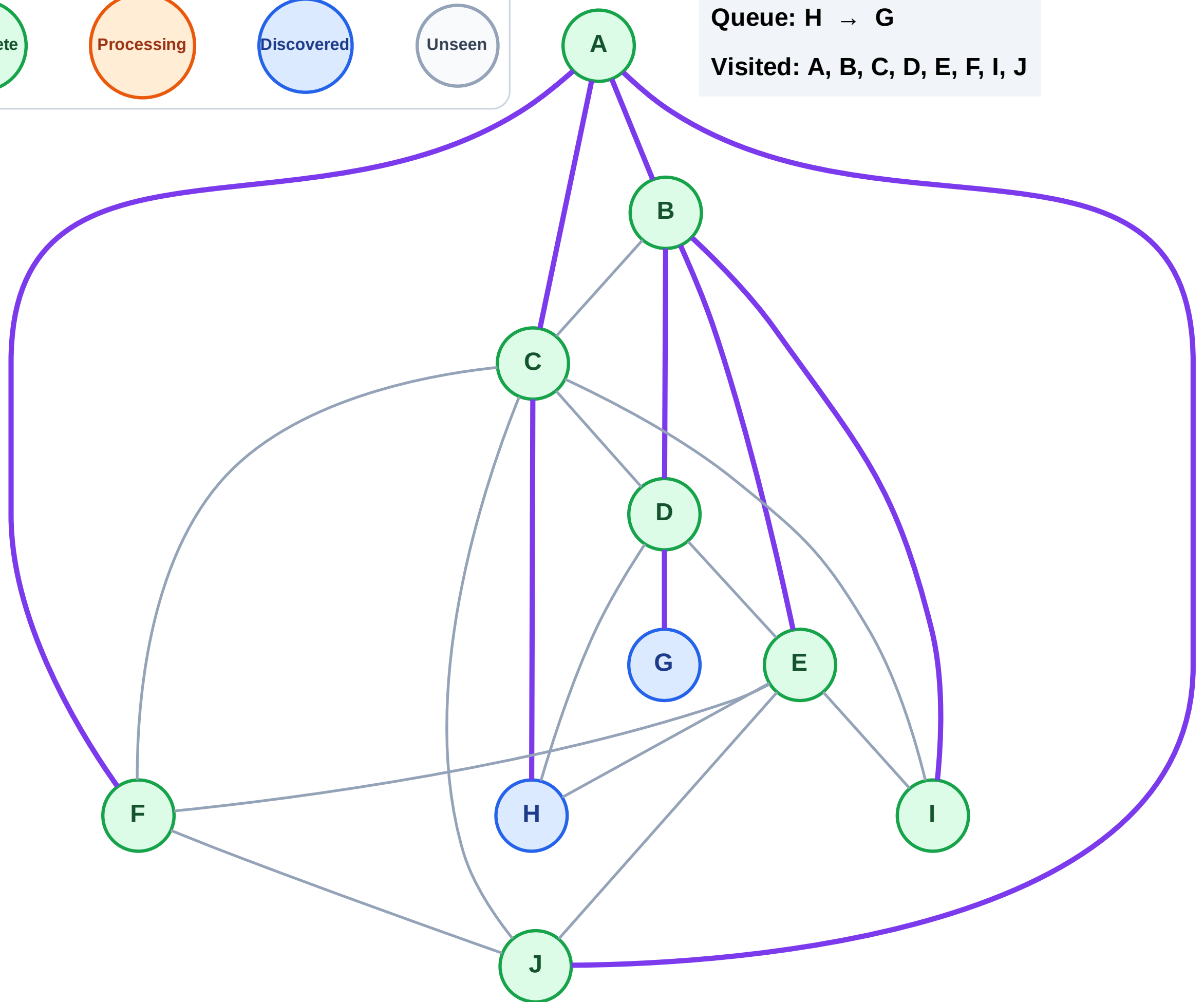
Discovered



Unseen

Queue: H → G

Visited: A, B, C, D, E, F, I, J



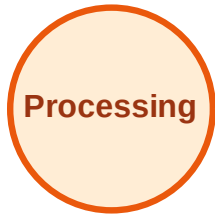
BFS Traversal

Step 18: Process node H

Legend



Complete



Processing



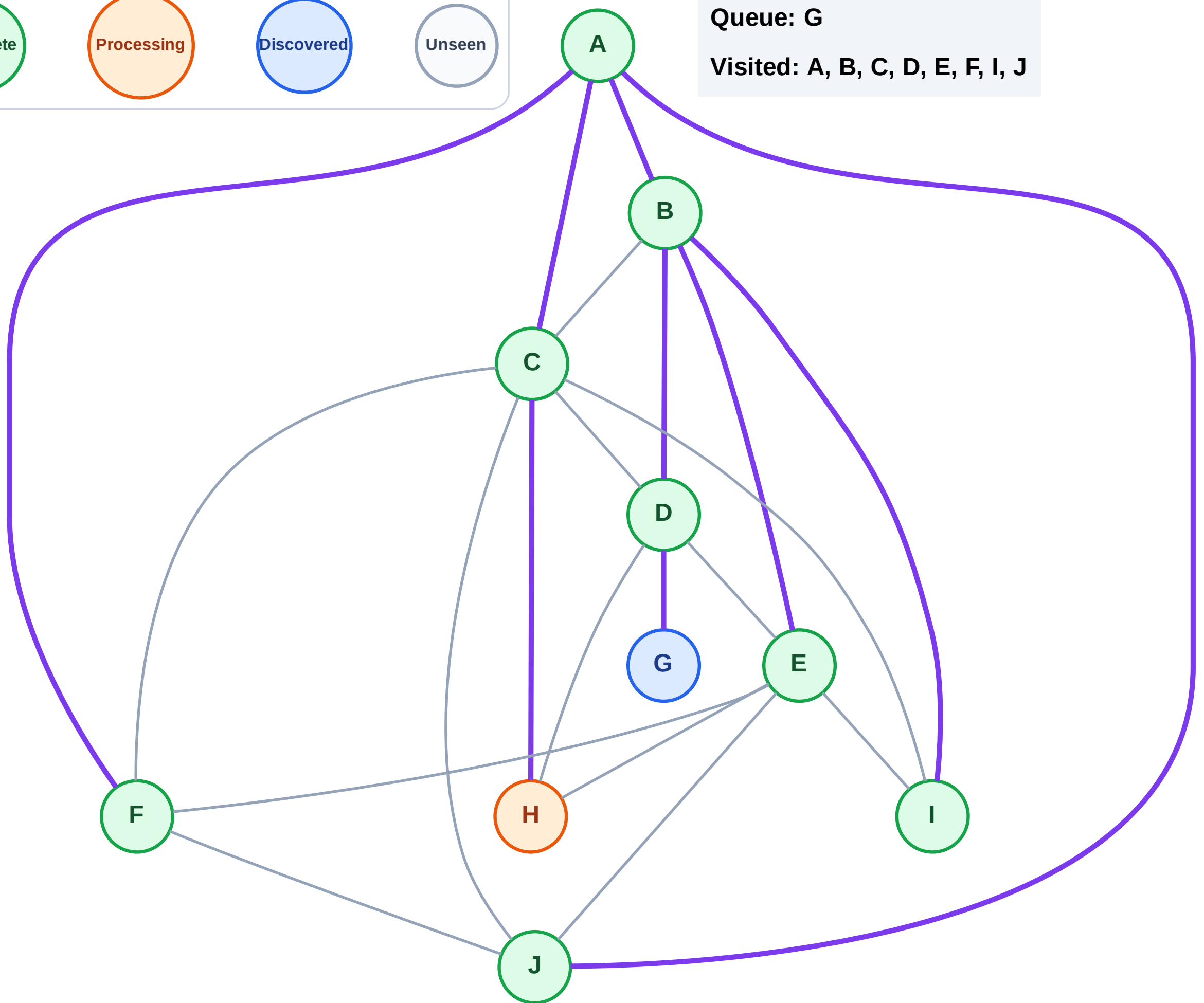
Discovered



Unseen

Queue: G

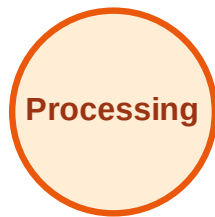
Visited: A, B, C, D, E, F, I, J



BFS Traversal

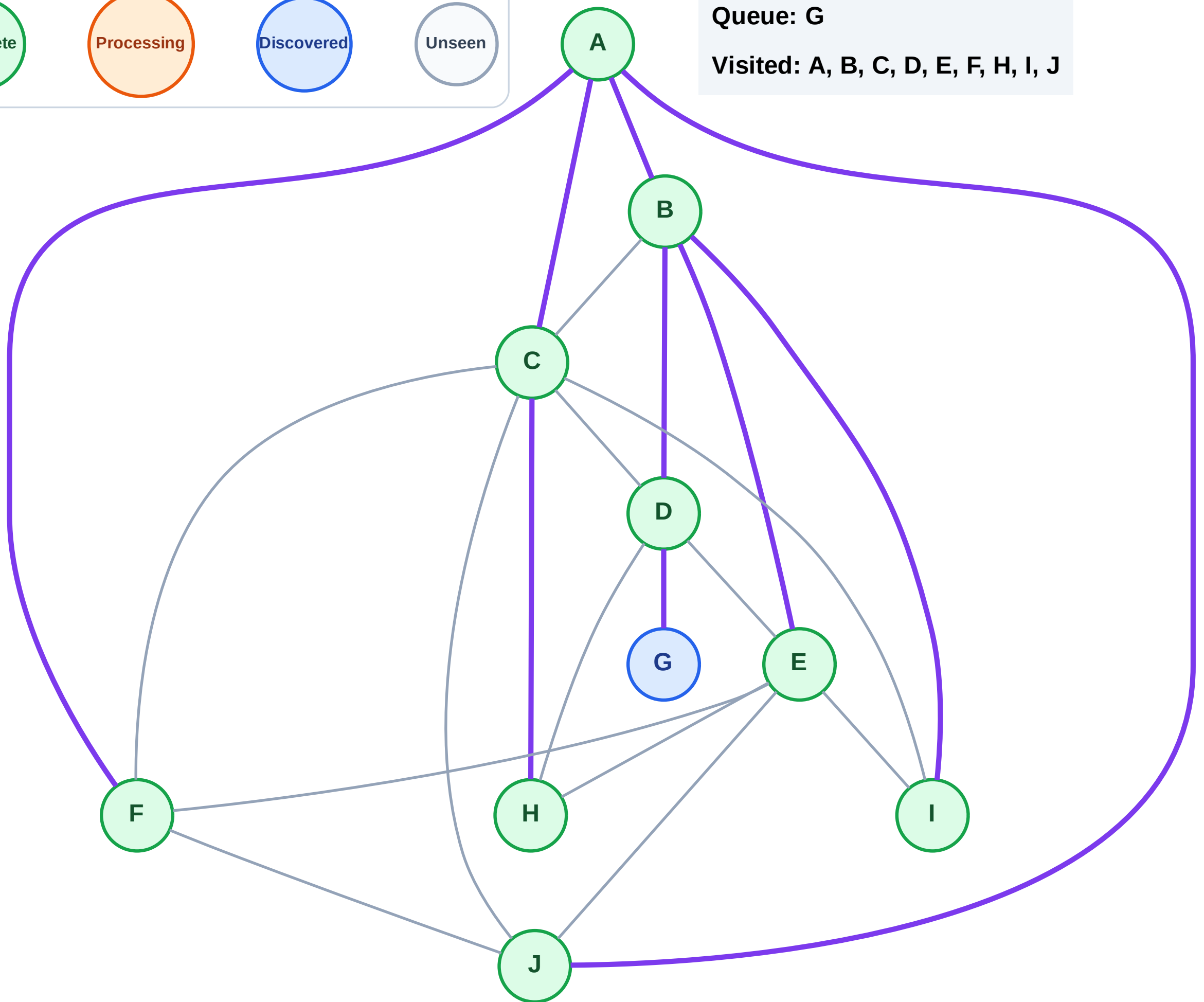
Step 19: No new neighbors from H

Legend



Queue: G

Visited: A, B, C, D, E, F, H, I, J



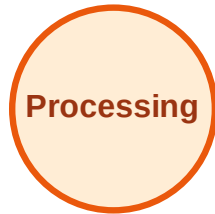
BFS Traversal

Step 20: Process node G

Legend



Complete



Processing



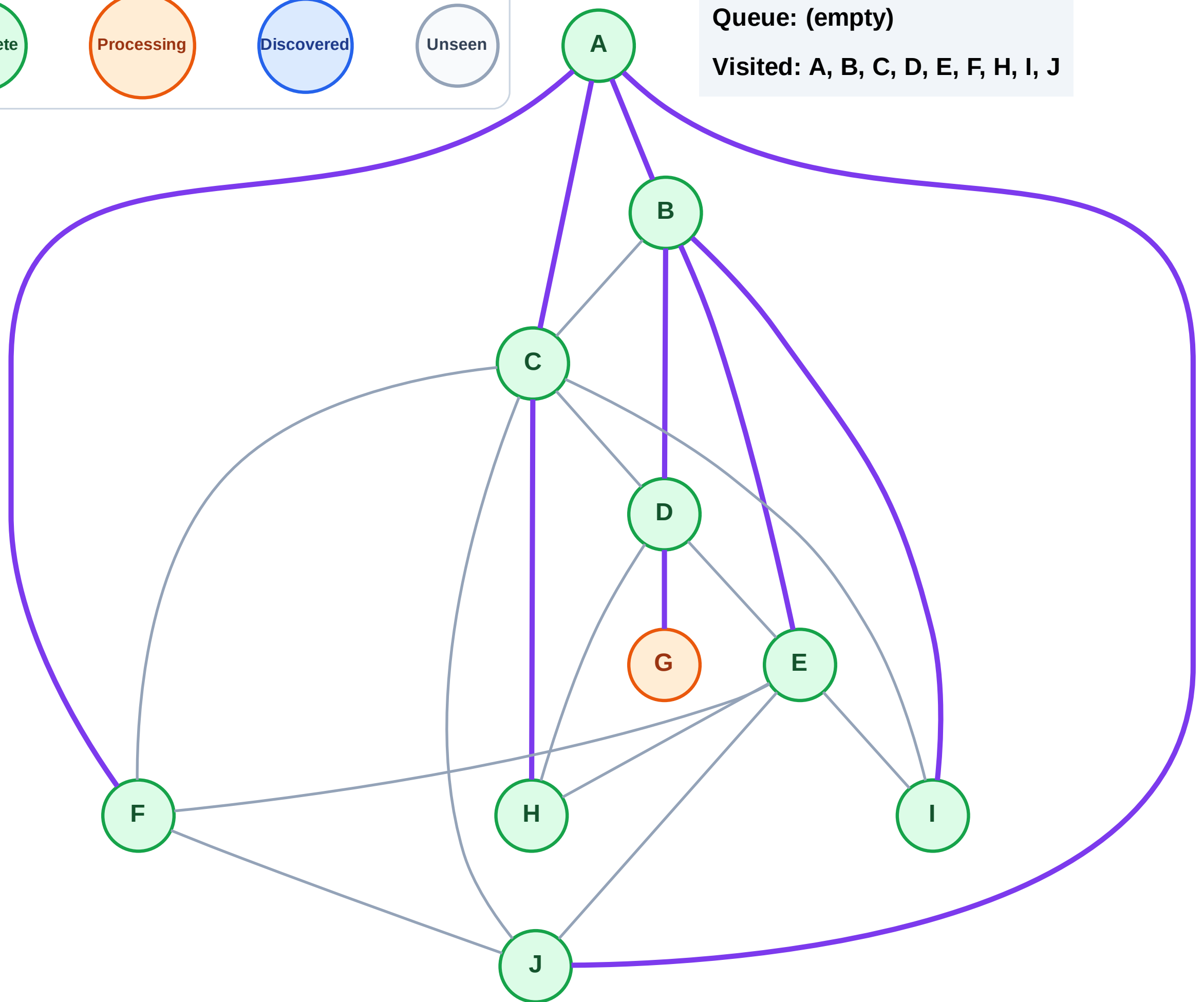
Discovered



Unseen

Queue: (empty)

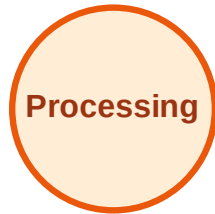
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

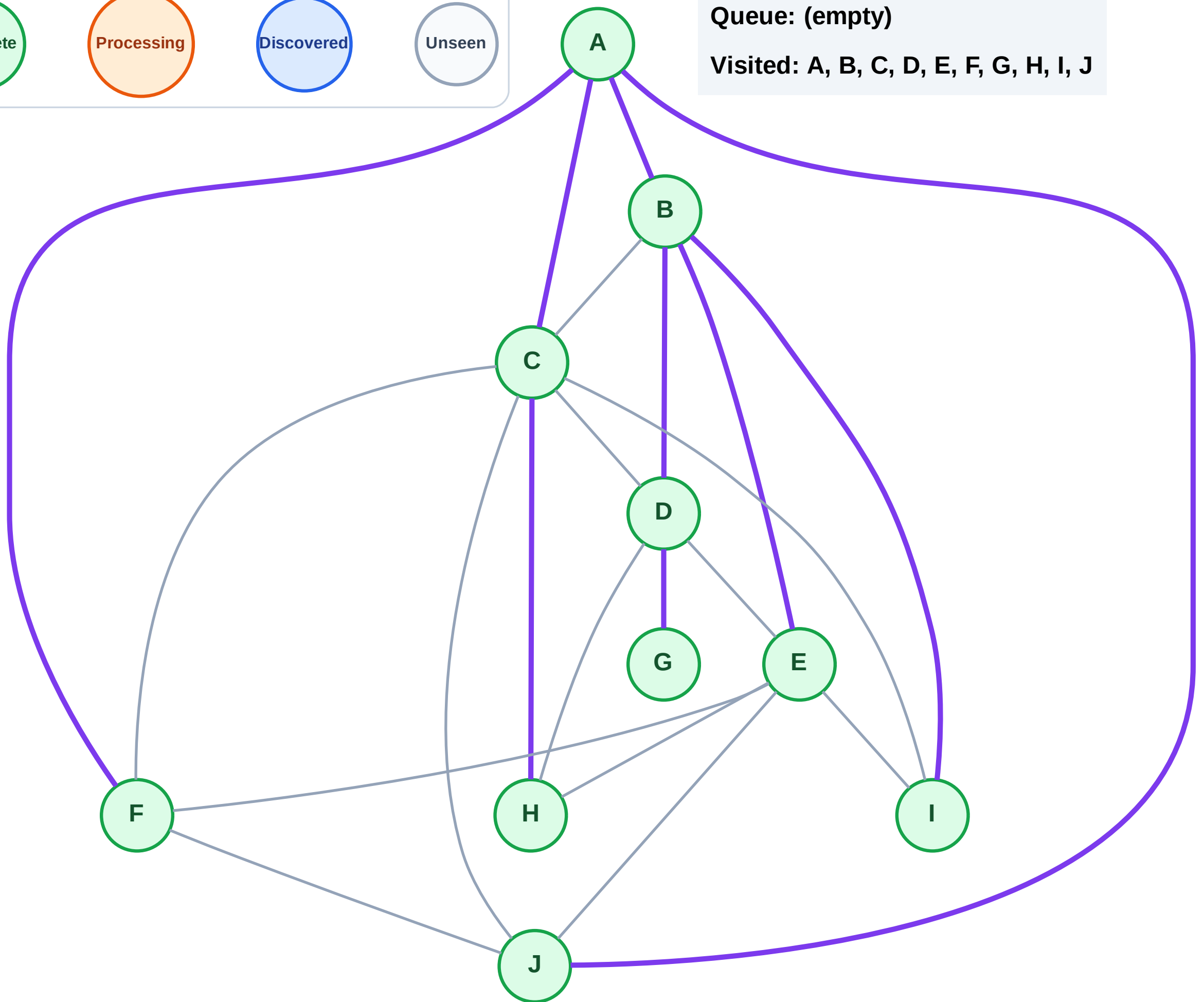
Step 21: No new neighbors from G

Legend



Queue: (empty)

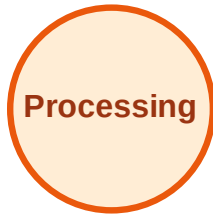
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

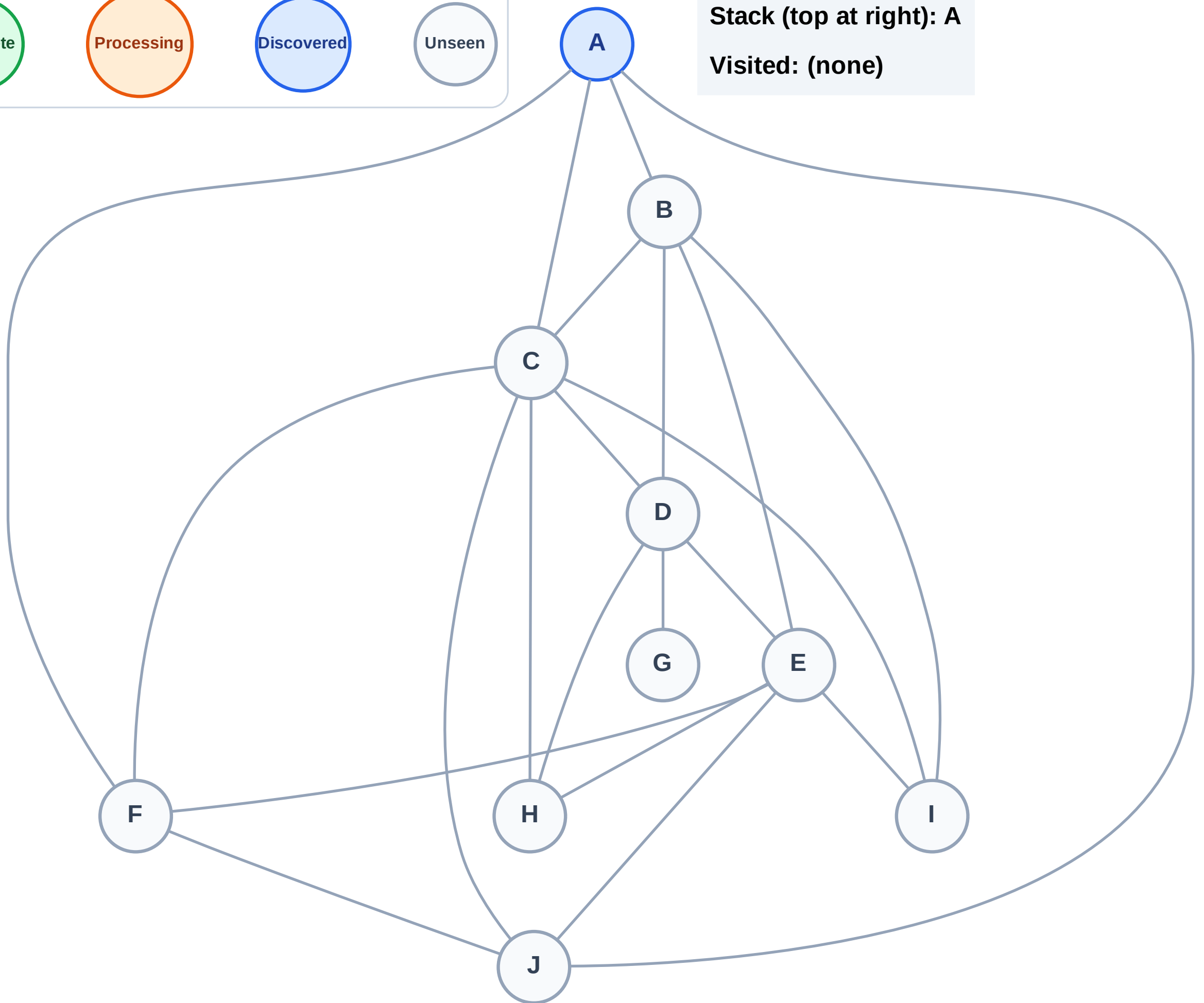
Step 1: Start at A

Legend



Stack (top at right): A

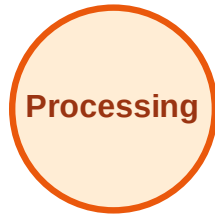
Visited: (none)



DFS Traversal

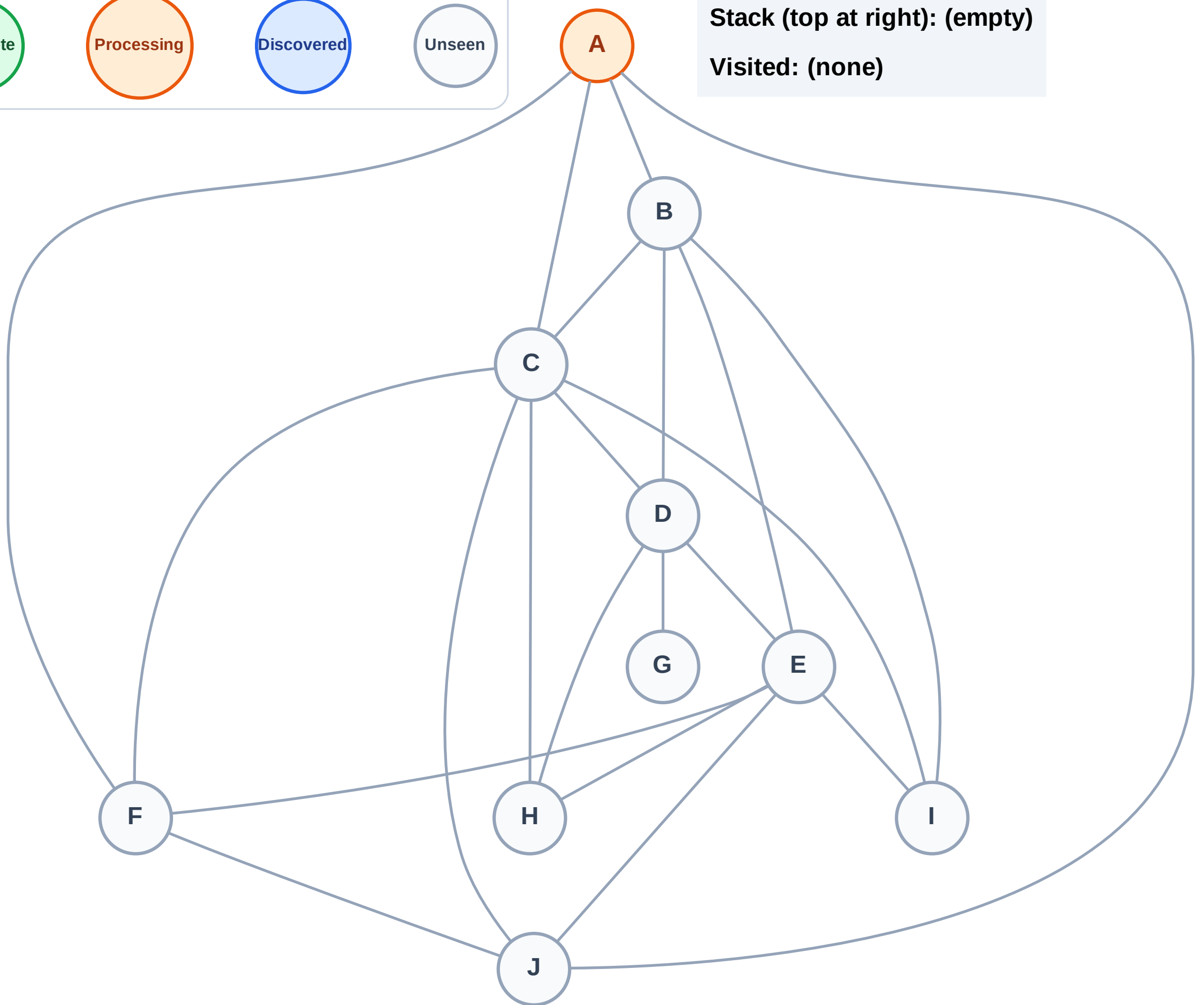
Step 2: Process node A

Legend



Stack (top at right): (empty)

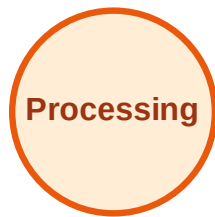
Visited: (none)



DFS Traversal

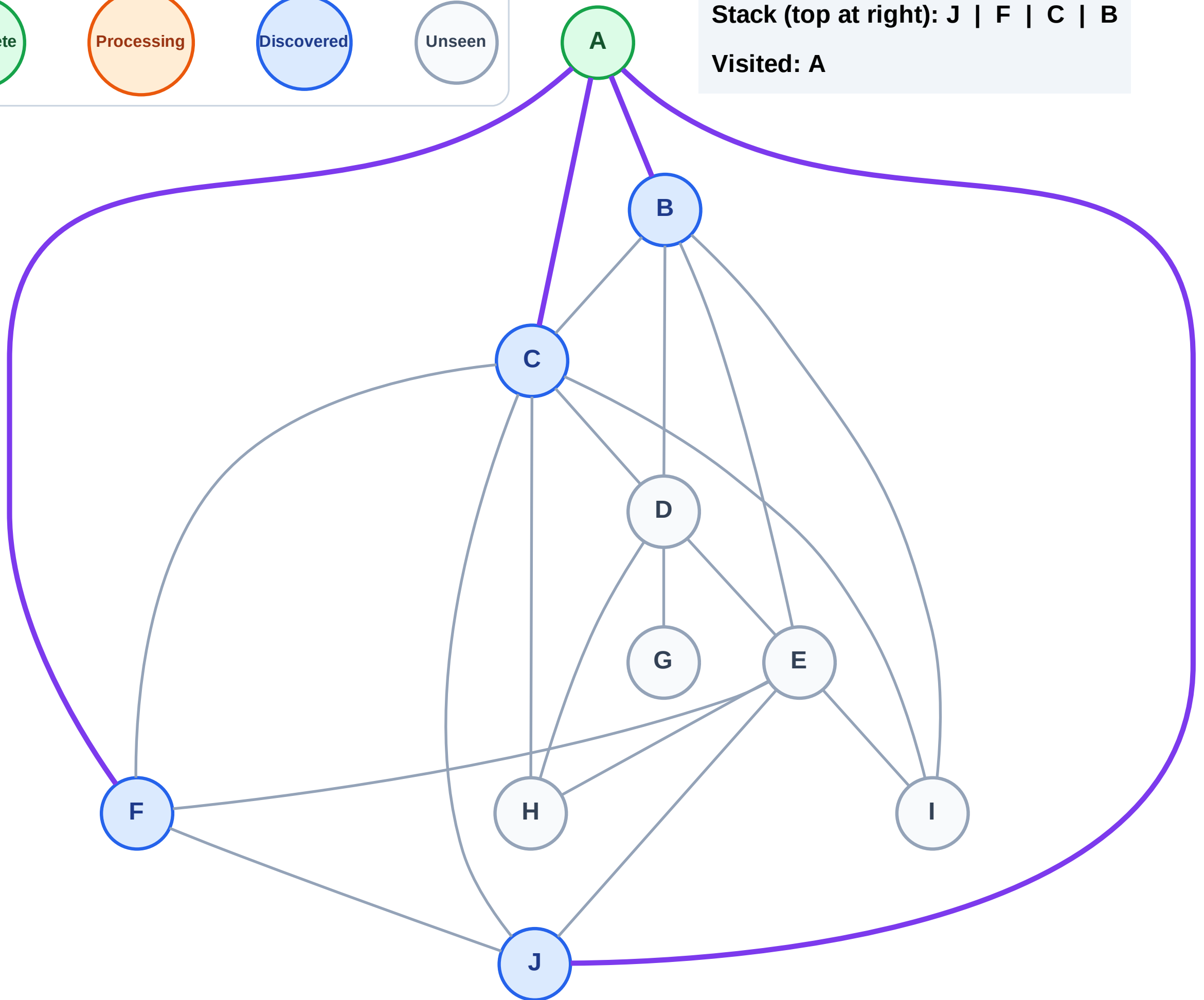
Step 3: Discover J, F, C, B from A

Legend



Stack (top at right): J | F | C | B

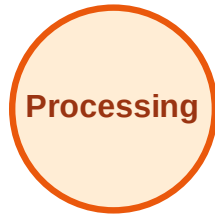
Visited: A



DFS Traversal

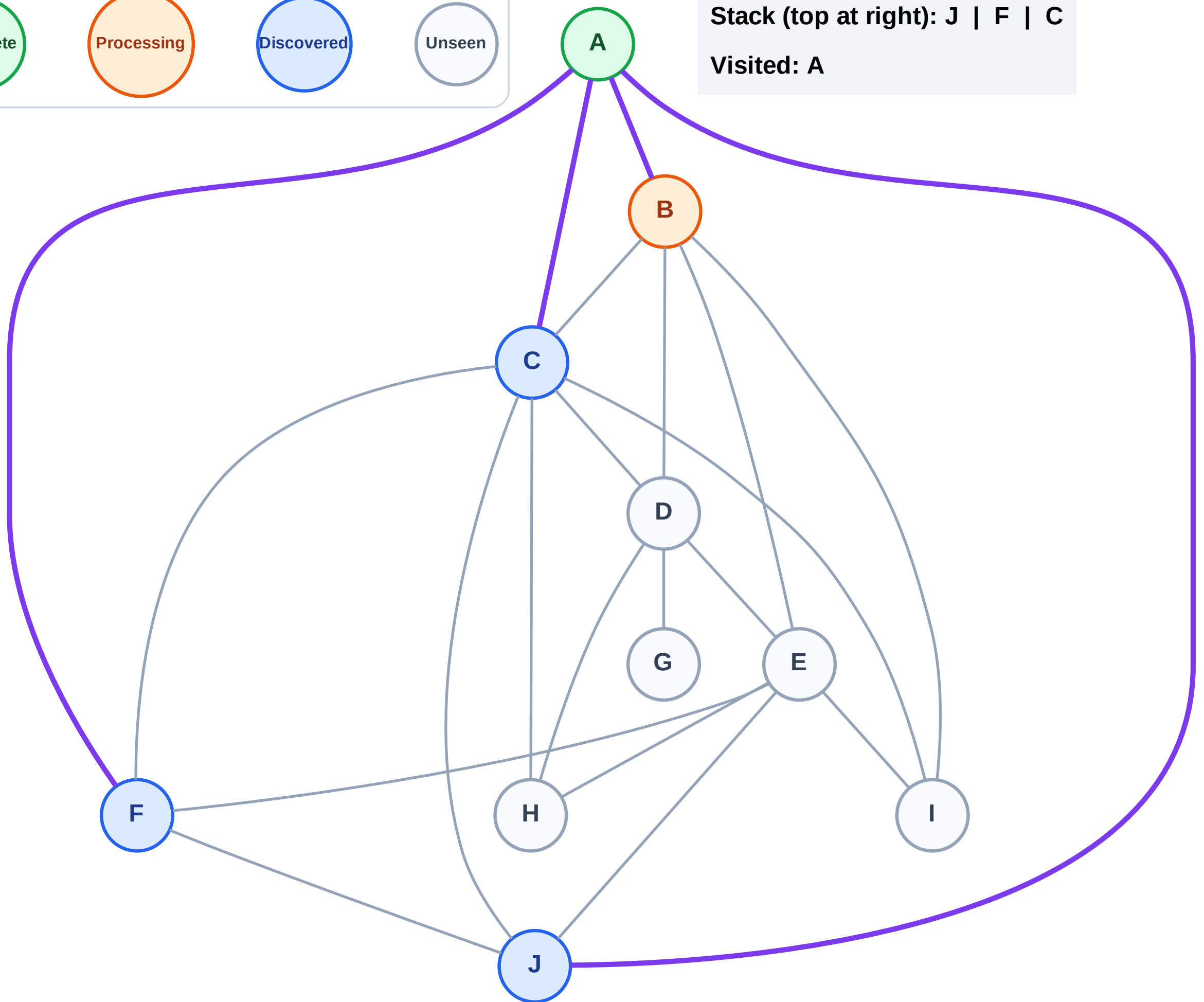
Step 4: Process node B

Legend



Stack (top at right): J | F | C

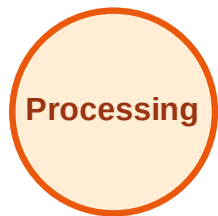
Visited: A



DFS Traversal

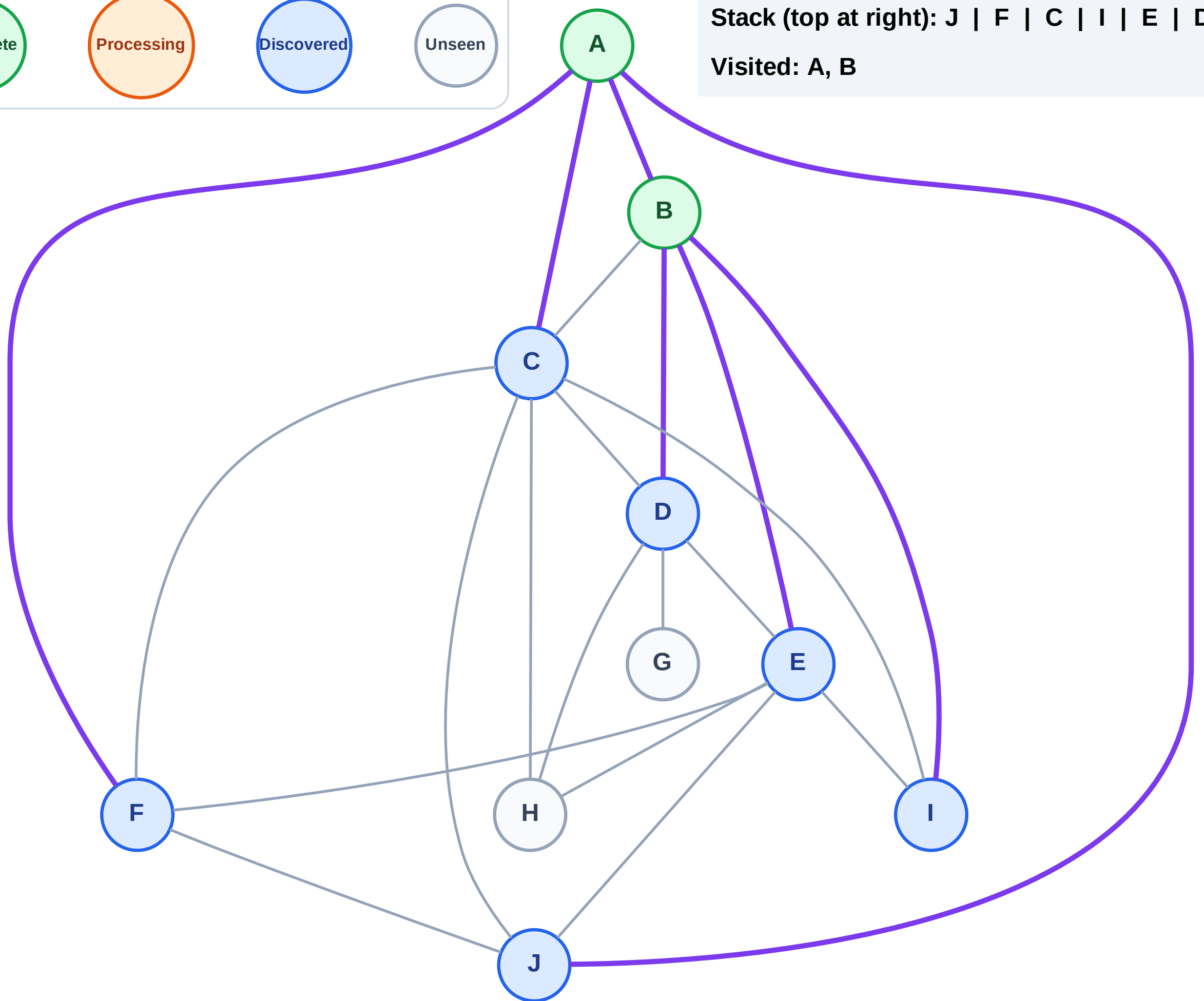
Step 5: Discover I, E, D from B

Legend



Stack (top at right): J | F | C | I | E | D

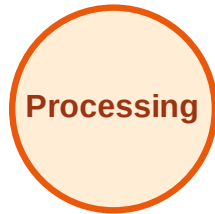
Visited: A, B



DFS Traversal

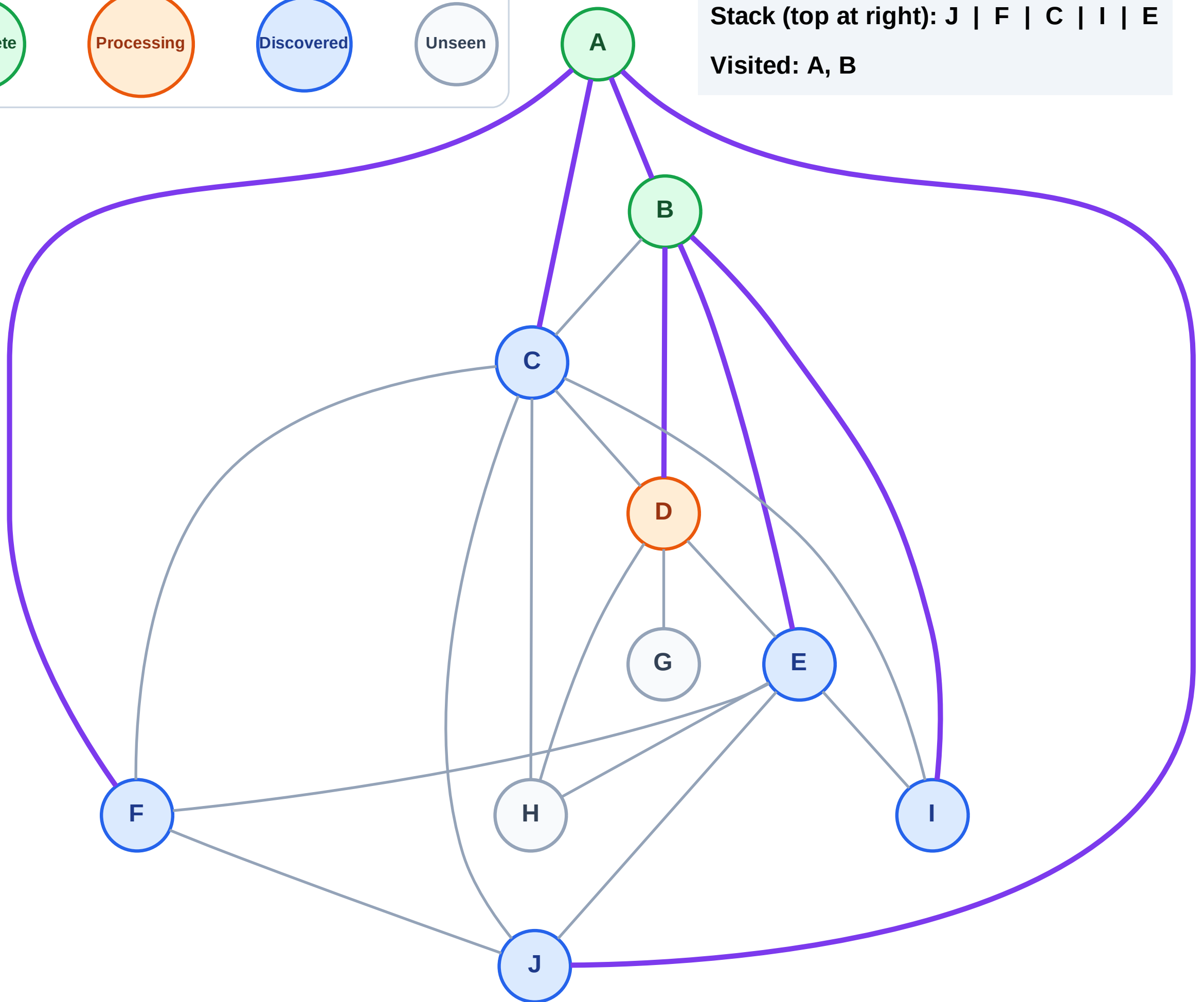
Step 6: Process node D

Legend



Stack (top at right): J | F | C | I | E

Visited: A, B



DFS Traversal

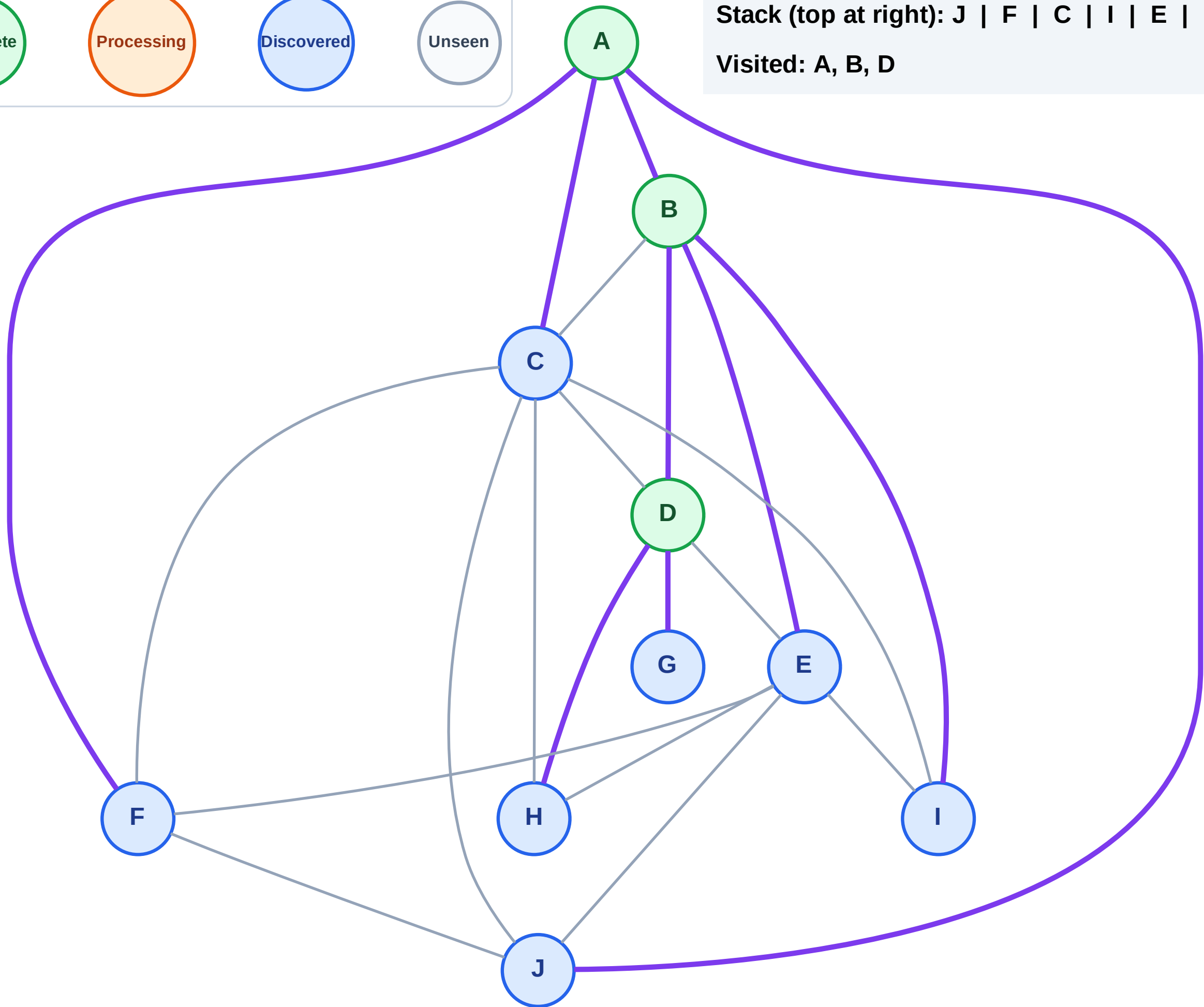
Step 7: Discover H, G from D

Legend



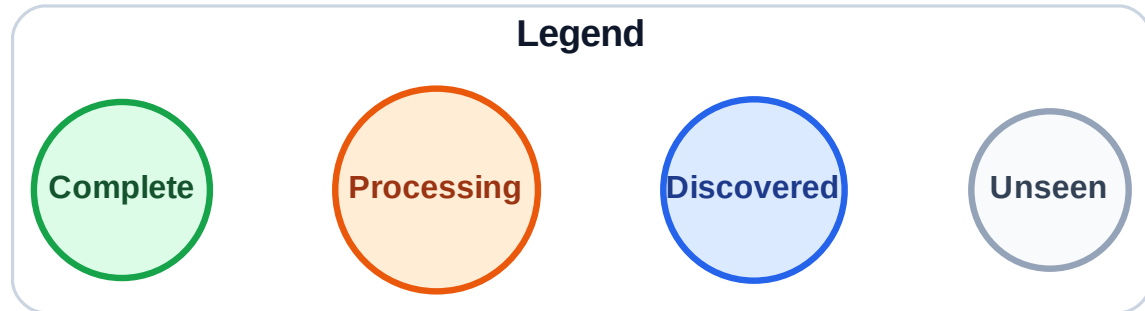
Stack (top at right): J | F | C | I | E | H | G

Visited: A, B, D



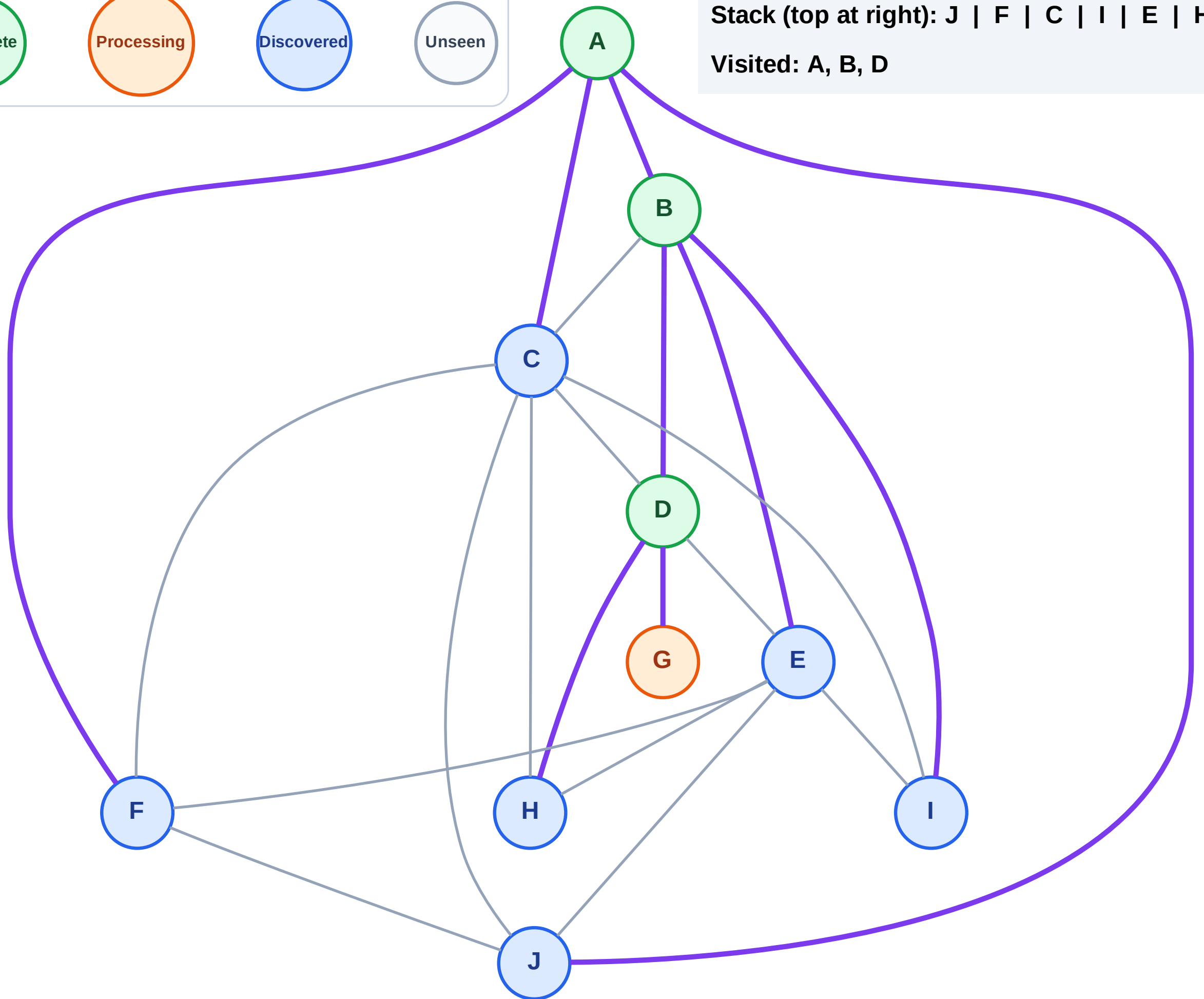
DFS Traversal

Step 8: Process node G



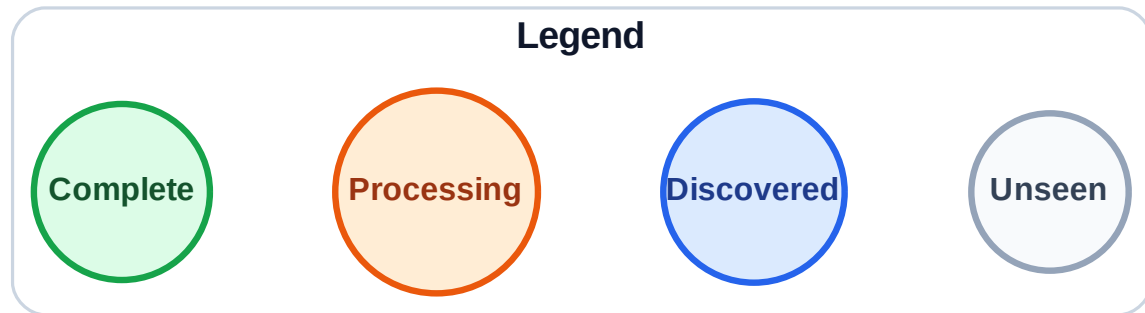
Stack (top at right): J | F | C | I | E | H

Visited: A, B, D



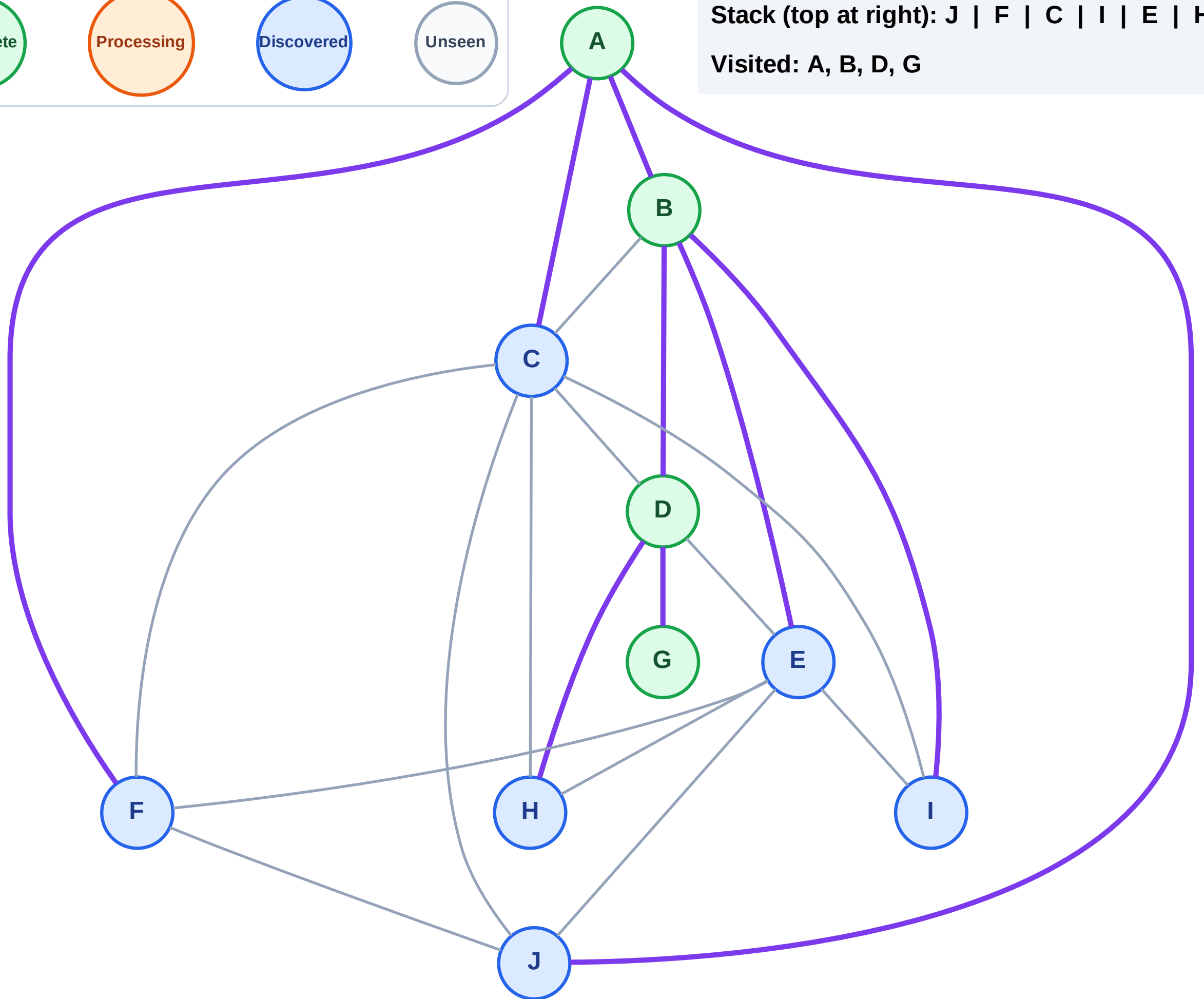
DFS Traversal

Step 9: No new neighbors from G



Stack (top at right): J | F | C | I | E | H

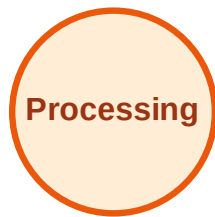
Visited: A, B, D, G



DFS Traversal

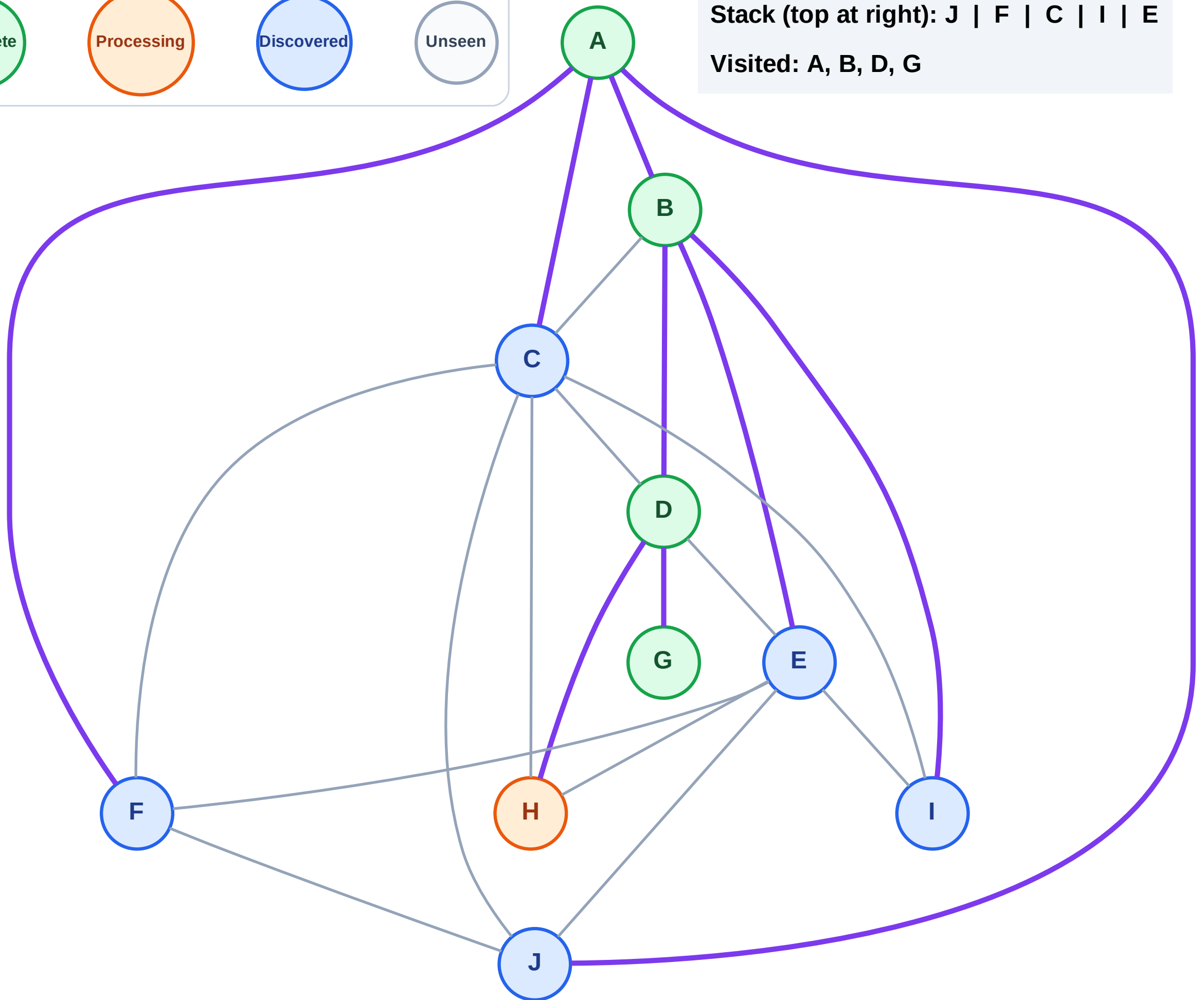
Step 10: Process node H

Legend



Stack (top at right): J | F | C | I | E

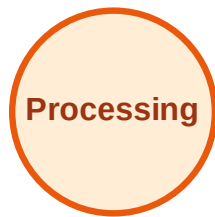
Visited: A, B, D, G



DFS Traversal

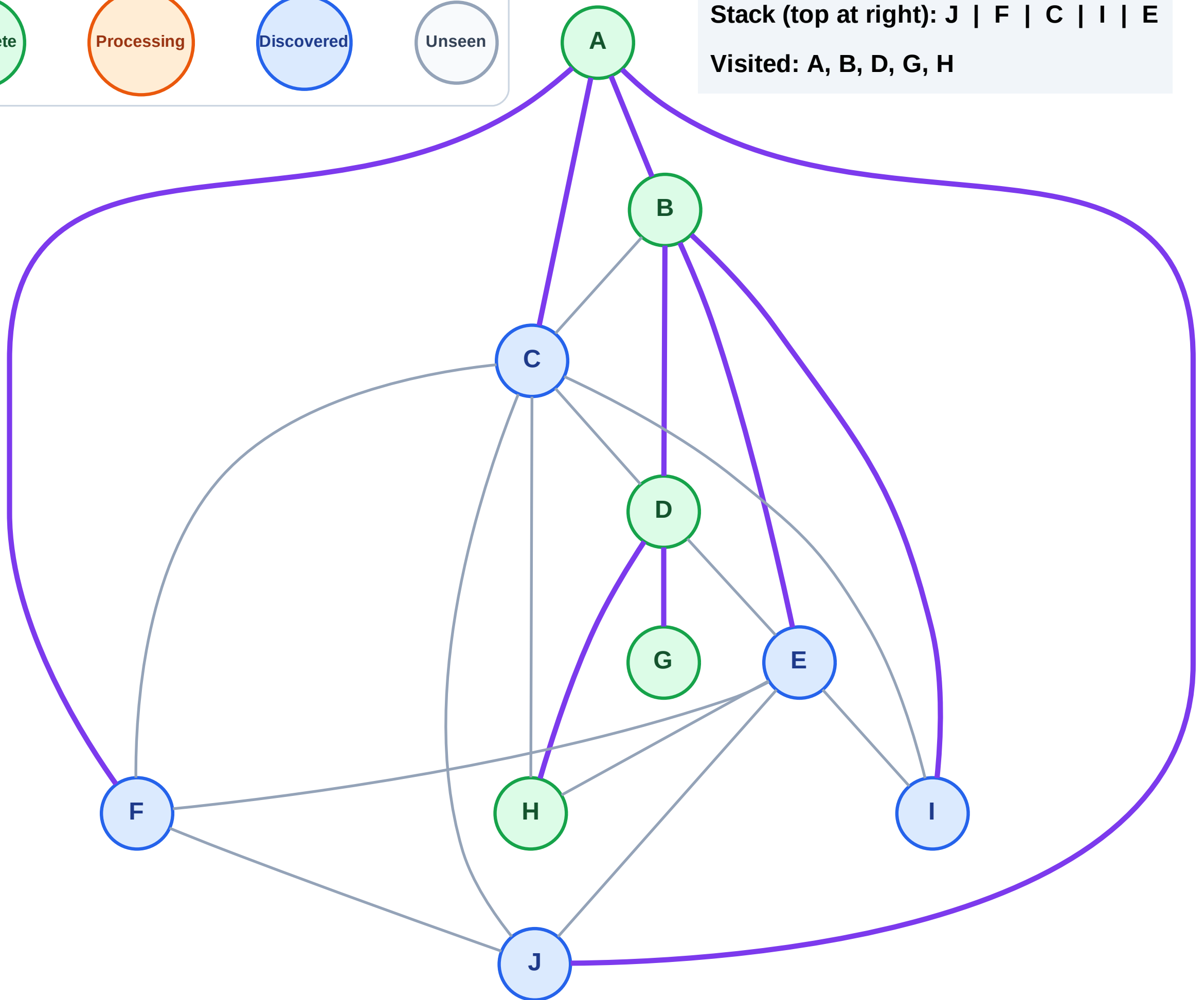
Step 11: No new neighbors from H

Legend



Stack (top at right): J | F | C | I | E

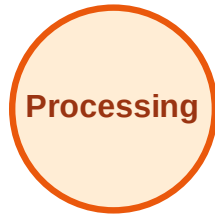
Visited: A, B, D, G, H



DFS Traversal

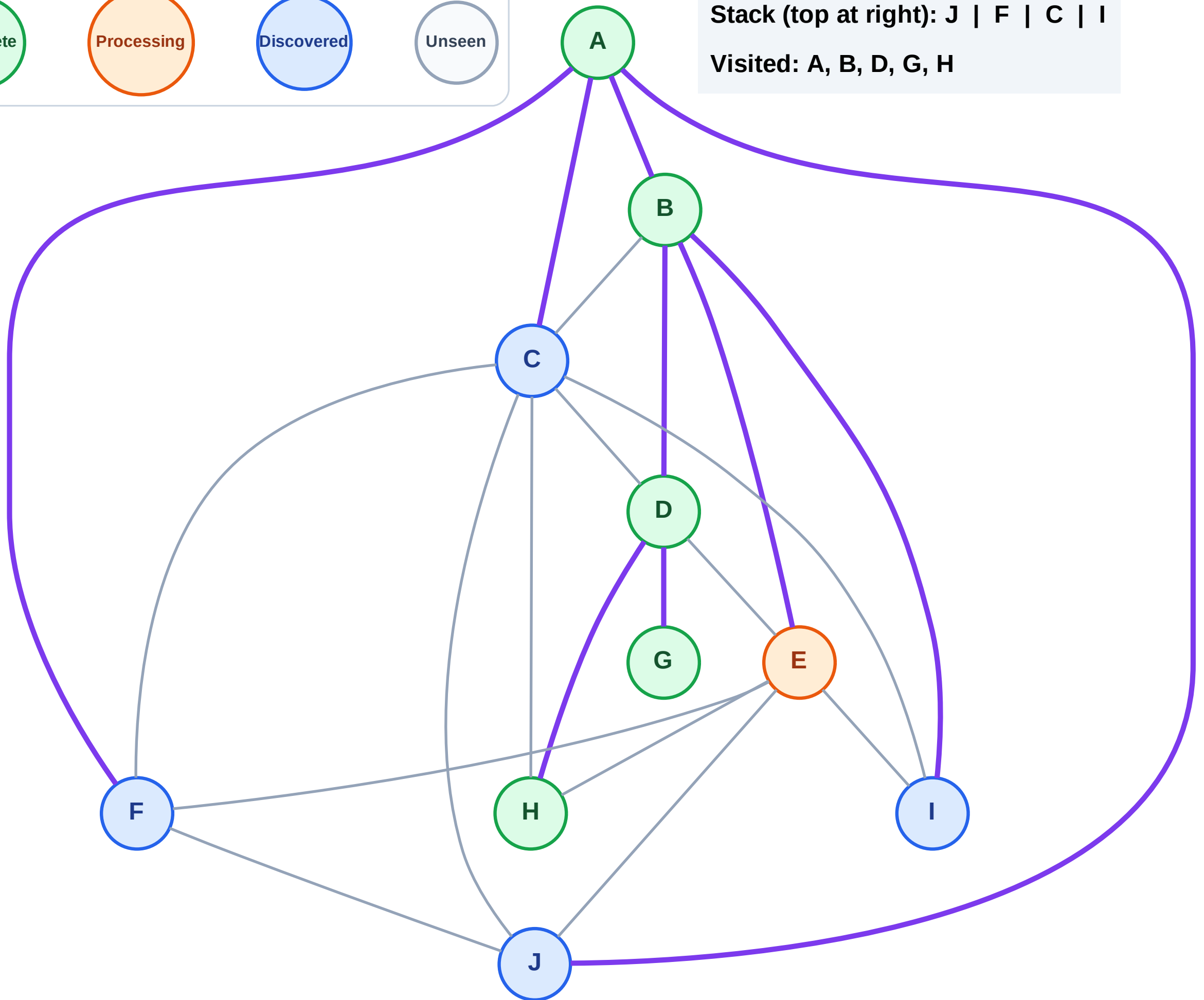
Step 12: Process node E

Legend



Stack (top at right): J | F | C | I

Visited: A, B, D, G, H

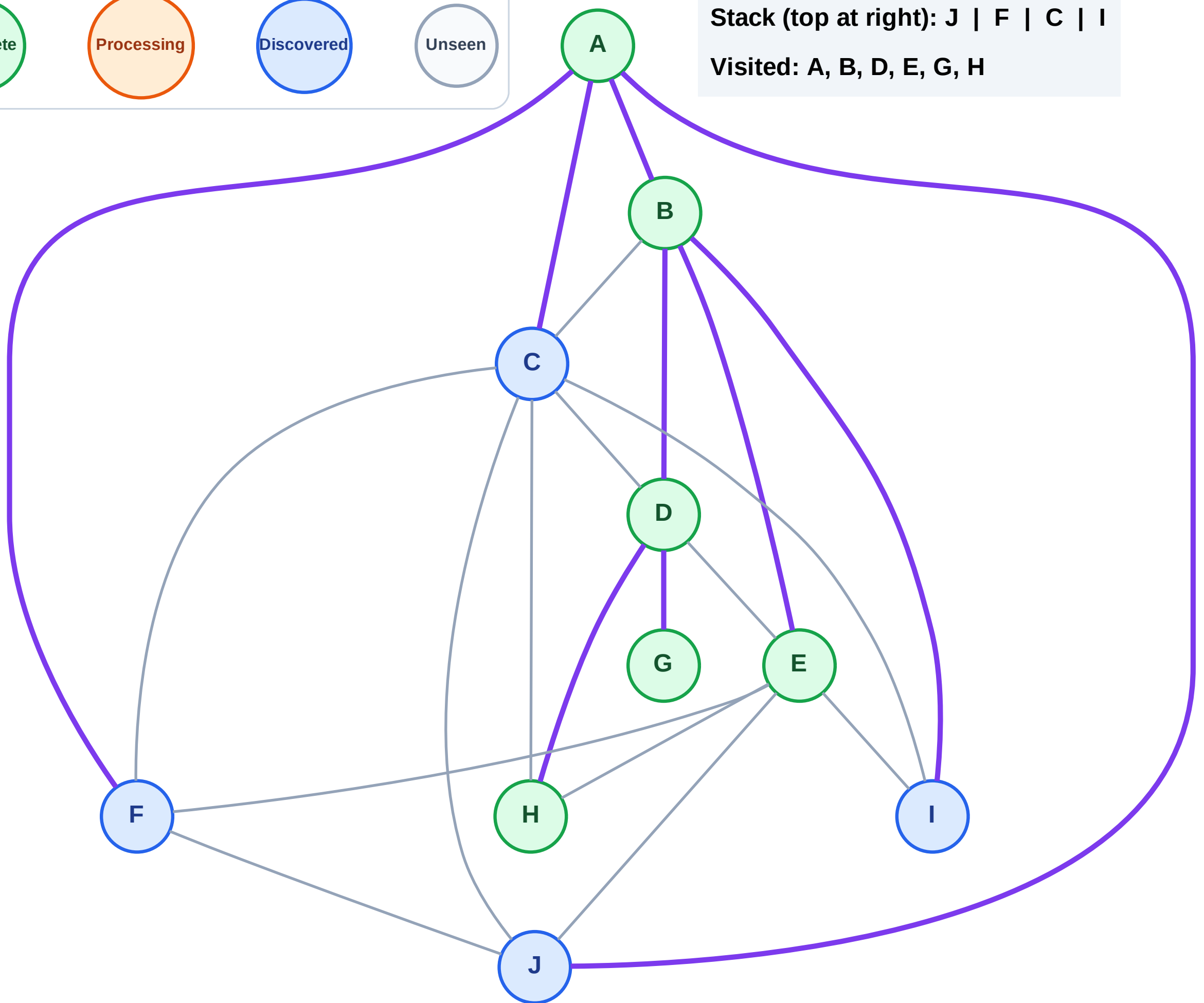


Step 13: No new neighbors from E

Legend

Unseen

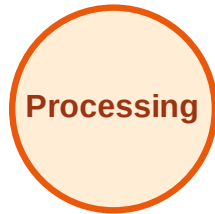
Visited: A, B, D, E, G, H



DFS Traversal

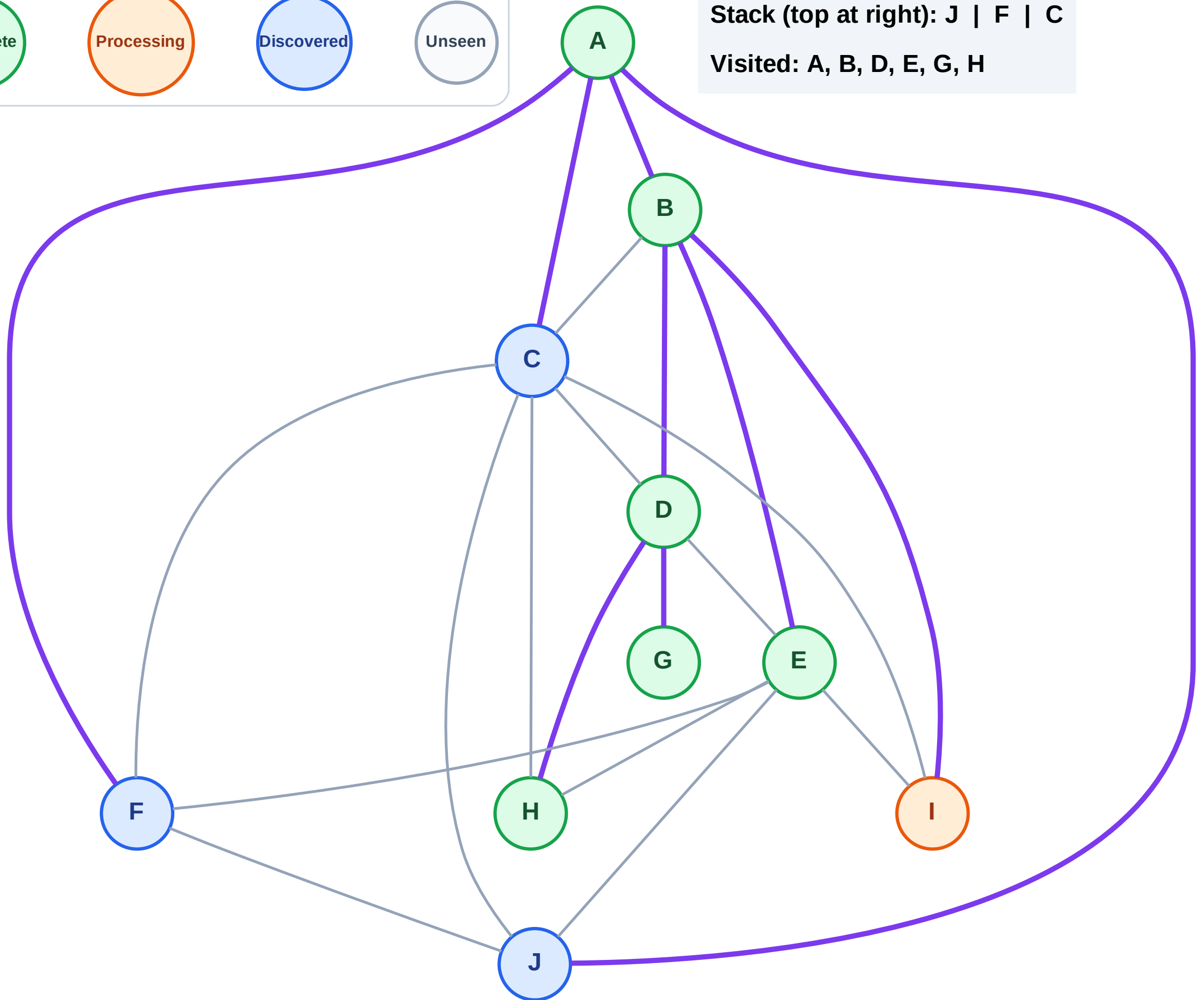
Step 14: Process node I

Legend



Stack (top at right): J | F | C

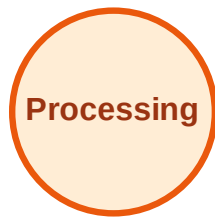
Visited: A, B, D, E, G, H



DFS Traversal

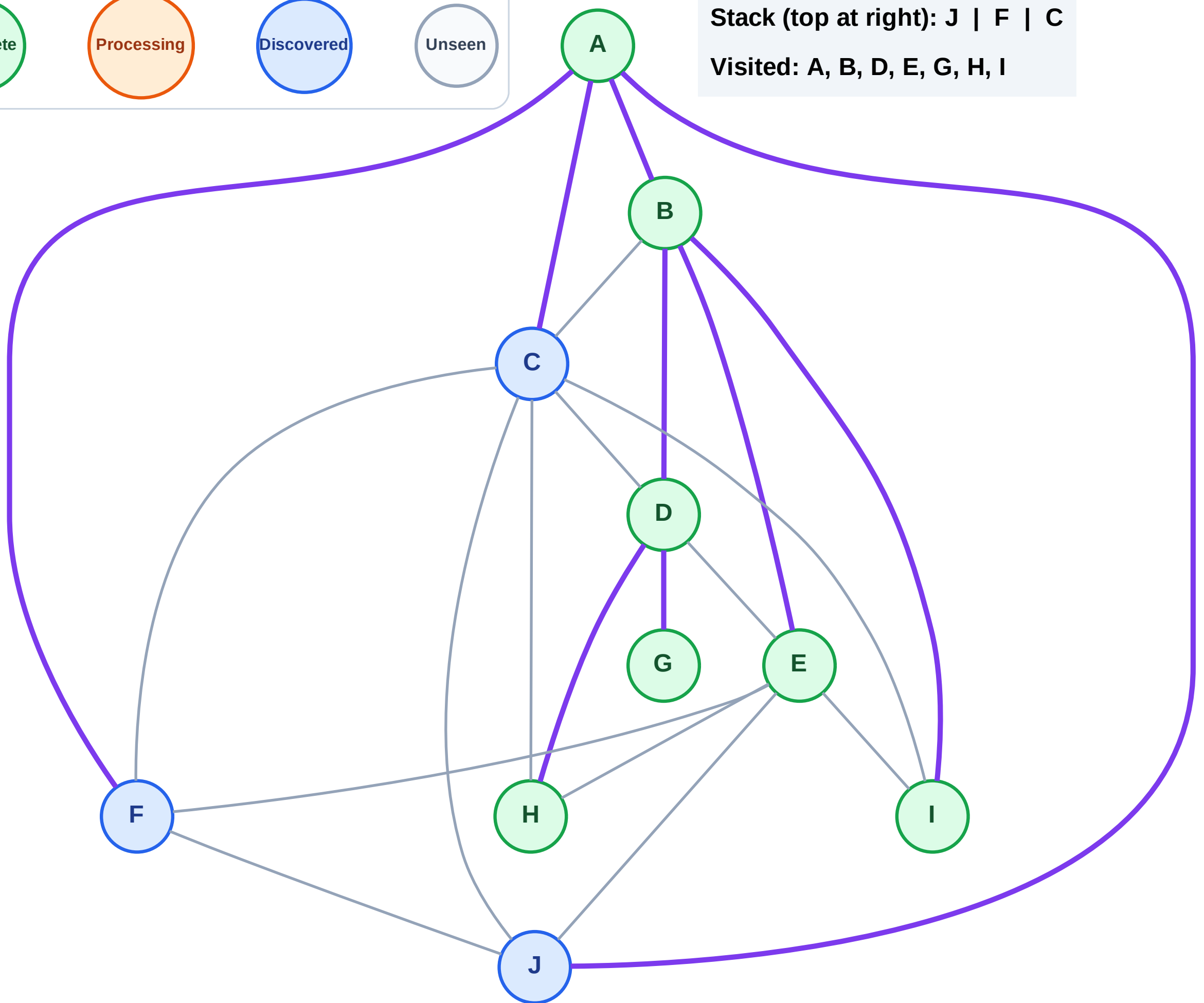
Step 15: No new neighbors from I

Legend



Stack (top at right): J | F | C

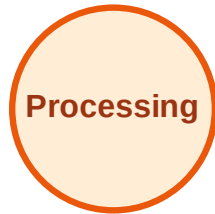
Visited: A, B, D, E, G, H, I



DFS Traversal

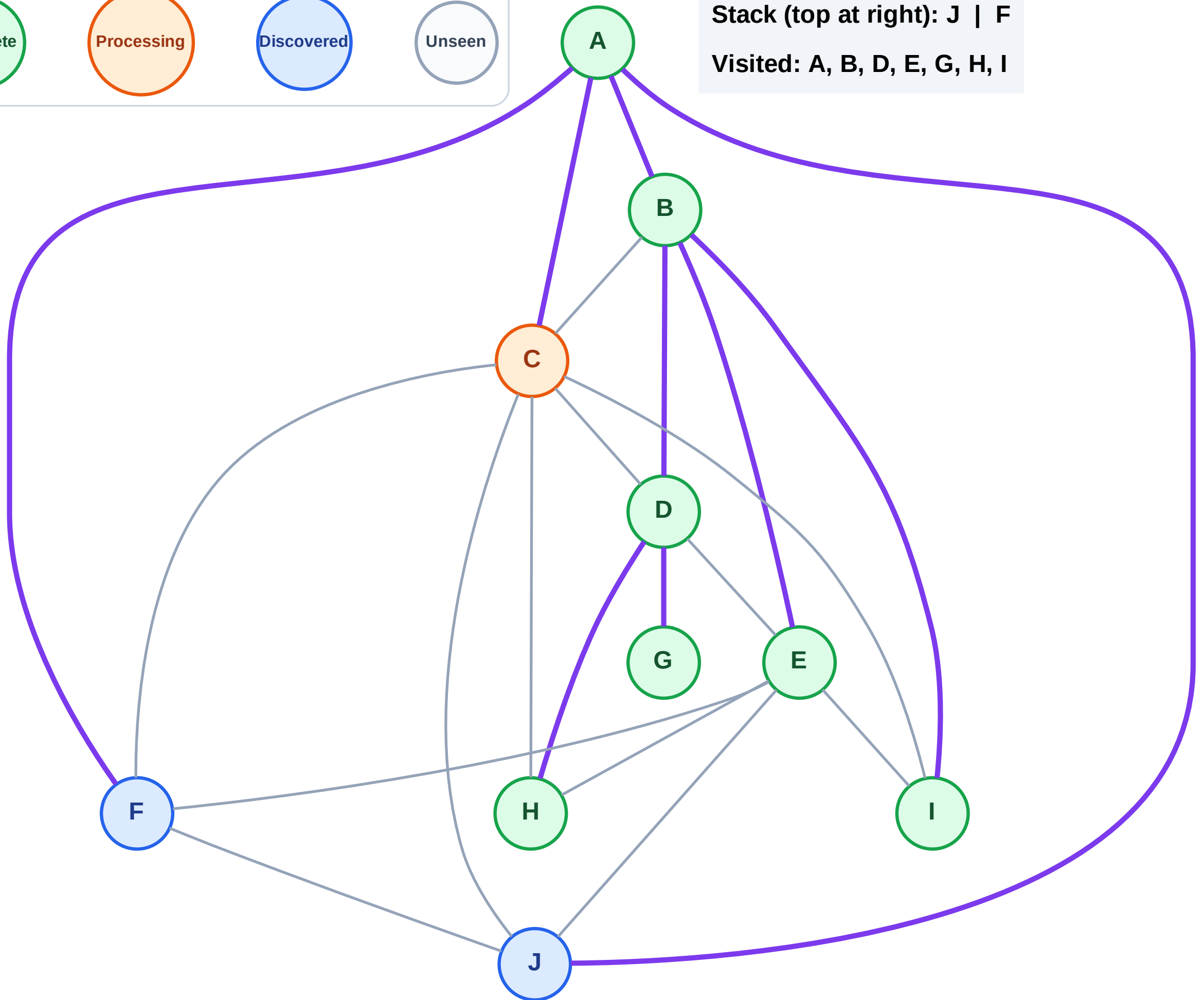
Step 16: Process node C

Legend



Stack (top at right): J | F

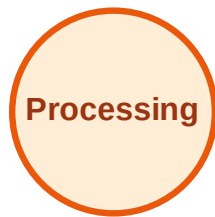
Visited: A, B, D, E, G, H, I



DFS Traversal

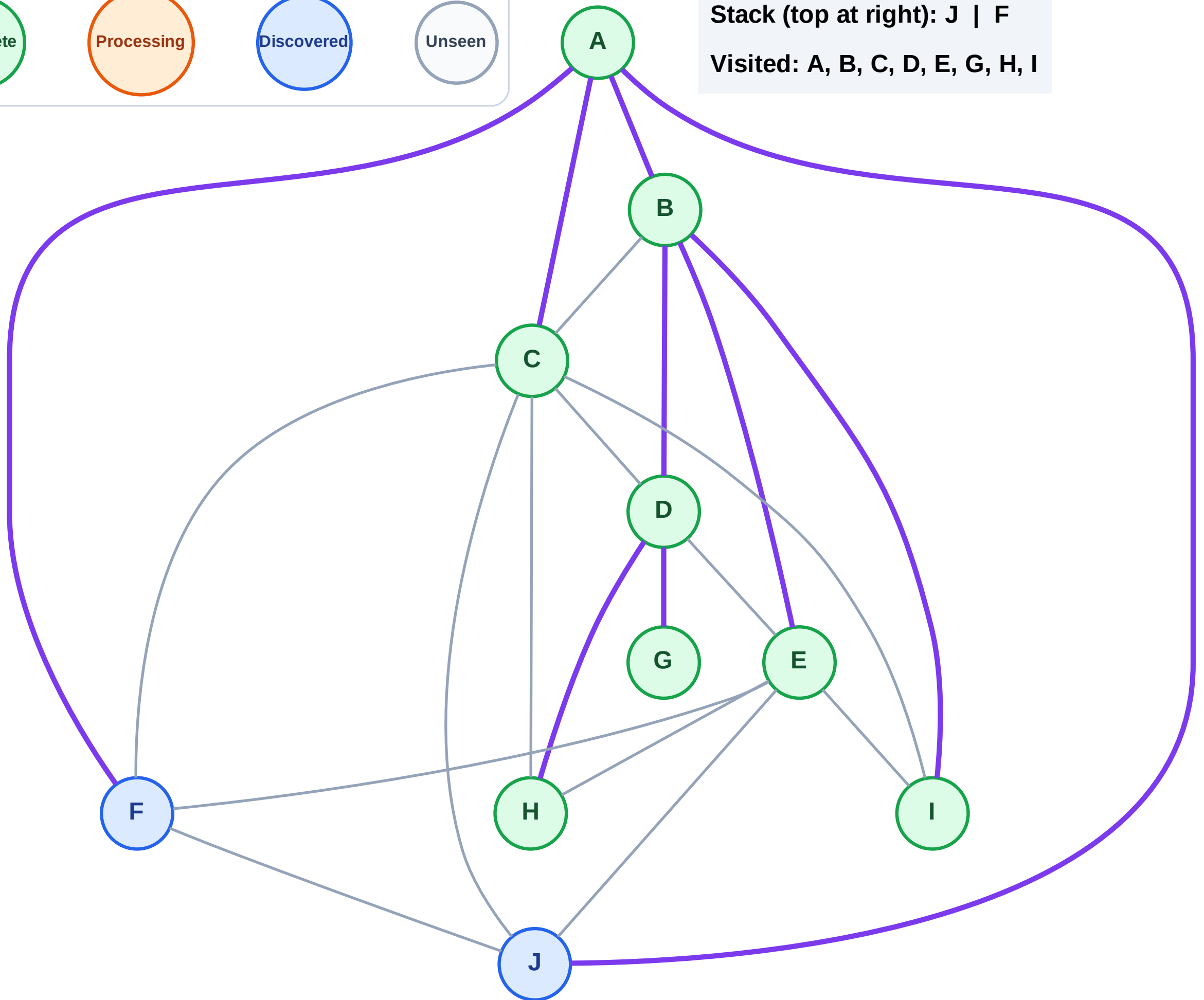
Step 17: No new neighbors from C

Legend



Stack (top at right): J | F

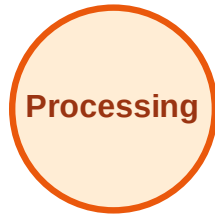
Visited: A, B, C, D, E, G, H, I



DFS Traversal

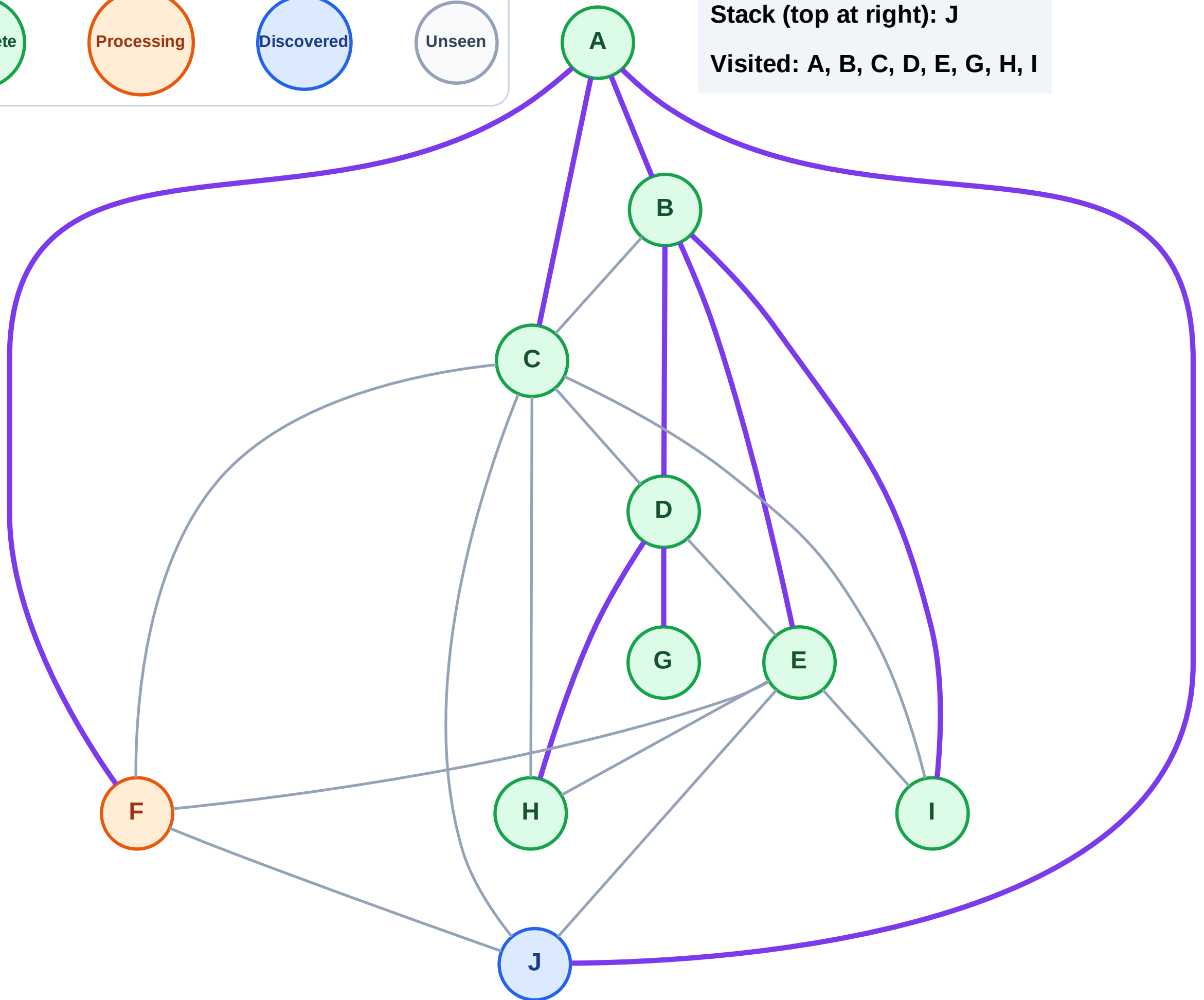
Step 18: Process node F

Legend



Stack (top at right): J

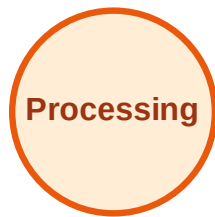
Visited: A, B, C, D, E, G, H, I



DFS Traversal

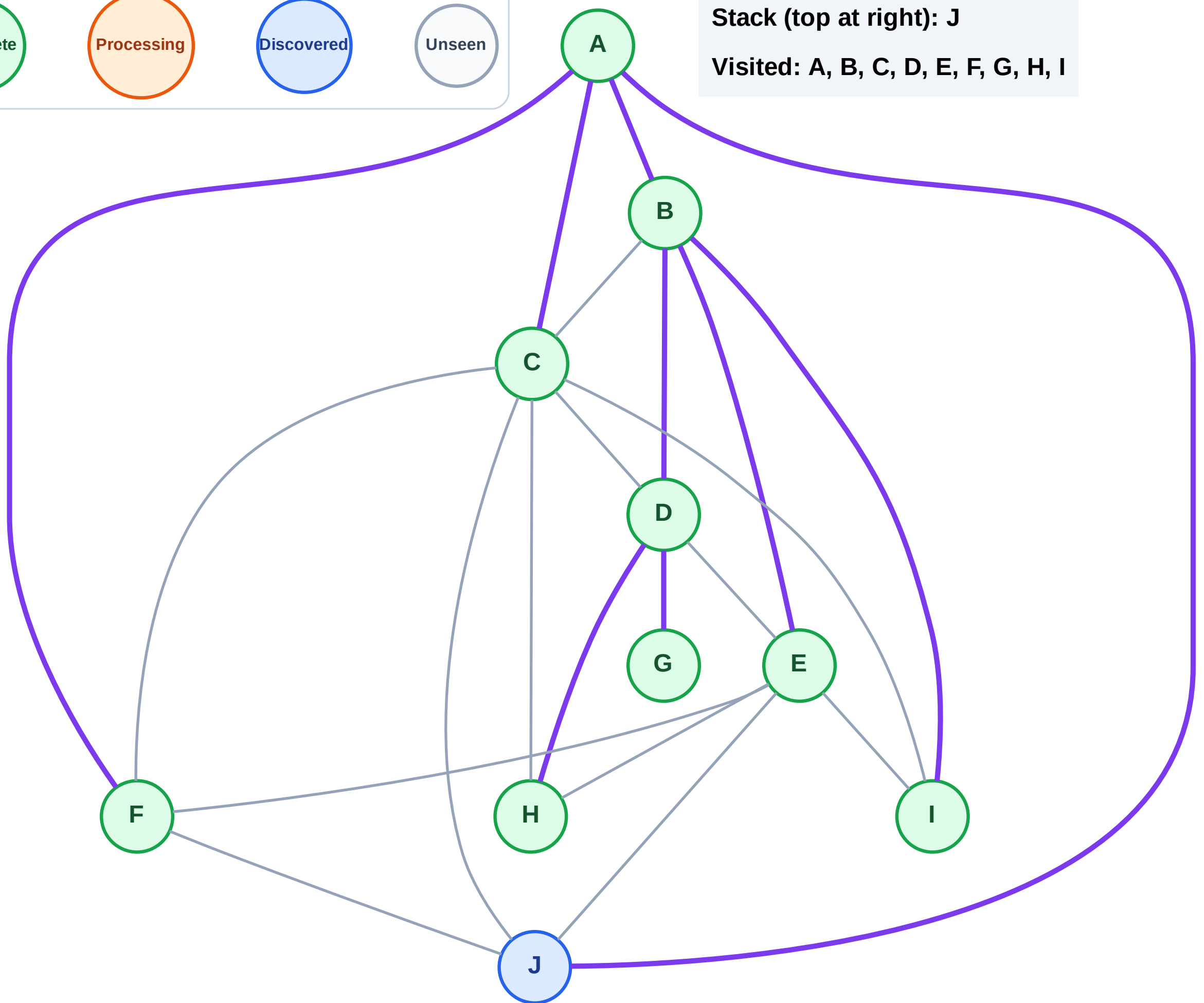
Step 19: No new neighbors from F

Legend



Stack (top at right): J

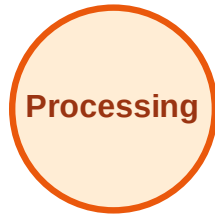
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

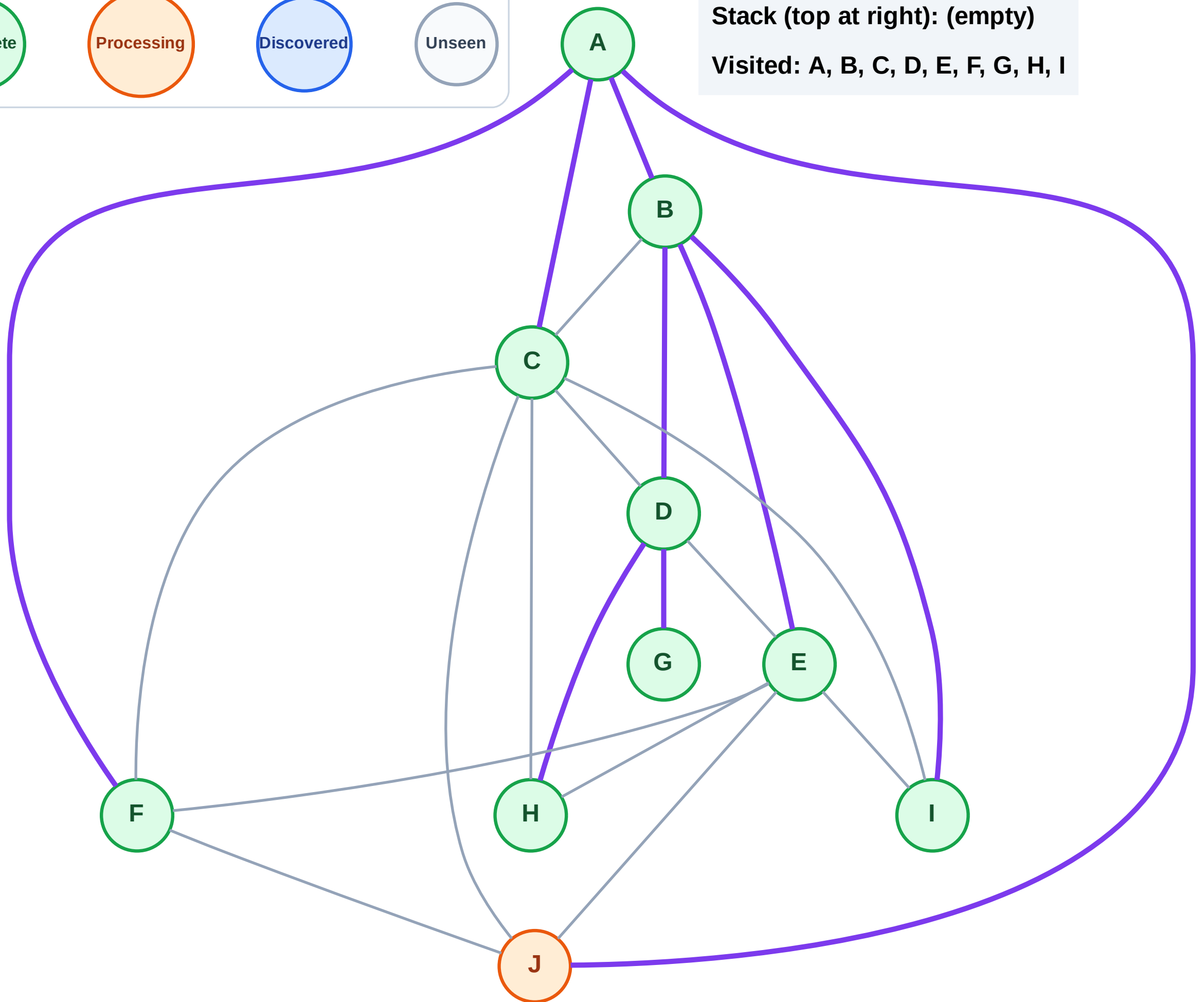
Step 20: Process node J

Legend



Stack (top at right): (empty)

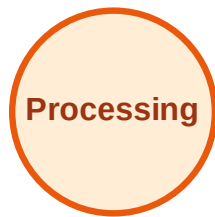
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

